A Categorical Viability-Weighted Curvature Framework: From Autopoietic Sets to Non-Abelian Holonomy

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Abstract. A stratified Fisher–viability transport framework is developed whose curvature predicts leading-order policy hysteresis on constant-active-set strata. The framework rests on the Stratified Autopoietic Viability Geometric System (SAVGS), a five-component object (smooth control manifold B, policy bundle $\pi : E \to B$ with fiber $\mathcal{P}$, strict viability margin ε, endogenous maintenance graph Γ, and a 2-categorical span gluing constraint-switching strata) whose geometry is a stratified G_C-reduced principal bundle with structure group $G_C \in \{O(r), SO(r), \mathrm{CO}(r)\}$ selected by the policy-fiber geometry ($O(r)$ from Fisher metric; $SO(r)$ if oriented; $\mathrm{CO}(r)$ only with explicit Weyl scale). The viability-weighted curvature κ_V is constructed as the positive part of the directional derivative of a Bregman divergence evaluated at a smooth finite-code rate-distortion surrogate $r_{\tau,\beta,D}$ (the algorithmic version dist_D is upper- semicomputable but non-differentiable and serves only as a conjectural upper envelope). The framework is then typed into seven optic interfaces whose composite is a well-defined typed endo-optic; under an explicit Lipschitz contraction bound, the realized update has a unique fixed point by Banach's theorem. A Bregman–Hessian Noether correspondence (geodesic Lagrangian $L = \frac{1}{2} g_\phi(\dot{q}, \dot{q}) - U$ with affine Hessian isometry ξ satisfying $\mathcal{L}_\xi g_\phi = 0$) supplies the affine-invariant structure required for the gauge invariance of κ_V. A seven-claim falsification hierarchy is operationalized in the minimal $n = 3$ prototype (abelian, $\mathfrak{so}(2)$) and in the $n = 4$ non-abelian prototype (policy fiber $SO(3)$, $[\mathfrak{l}_z, \mathfrak{l}_x] = \mathfrak{l}_y$); the heavy-tail index of the fatigue noise is stress-tested across the Student–t tail-parameter axis. The main proposition states that on smooth constant-active-set strata, Fisher-minimal constraint-preserving adaptation defines a stratified connection whose viability-weighted curvature predicts leading-order policy hysteresis; whether that holonomy is fatal depends separately on viability margins, along-path disturbances, and regeneration of maintenance machinery. Five precise conjectures (global stratified holonomy; algorithmic upper envelope; filtered colimits in optic categories; heavy-tail 3/2 exponent; quantum self-reference by Zeno contraction) preserve the broader categorical, algorithmic, and quantum ambitions where their full proofs are out of reach. The core claims are numerically supported under stated tolerances; several extended claims are established under explicit additional hypotheses.

Keywords: viability theory, optic category, RAF sets, Fisher–Rao metric, CPTP channels, quantum Zeno effect, non-abelian holonomy, heavy-tailed noise, Banach fixed point, Bregman divergence, algorithmic rate-distortion, Noether correspondence.

1 Introduction

The notion that adaptation of a system to viability constraints can be unified across cognitive, biological, and physical domains has a long informal history in autopoiesis, viability theory [1], and information geometry [2, 3], but the precise categorical content of the unification has remained in flux. The dominant informal pattern is that viability- preserving systems, reflexively autocatalytic food-generated (RAF) reaction networks, predictive (Bayesian) belief updates, and quantum measurement schedules each admit a description in terms of a cost functional C whose stabilizer $\mathrm{Stab}(C)$ defines the structure group of the category of admissible updates. This pattern, however, has been expressed as a series of bridge-like correspondences without a single composition construction making the unification object a fixed point of a well-defined endofunctor.

This article assembles seven arcs of prior domain-unification work into a single endofunctor T on the optic category $\mathbf{Optic}(\mathcal{C})$ of Riley [4] and Brunerie et al. [5], where $\mathcal{C}$ is taken to be a category with finite limits. The composition $T = \mathcal{O}_7 \circ \cdots \circ \mathcal{O}_1$ of the seven optics is well-defined, associative, and unital; the unification object is the fixed point of T whenever T is contractive (Theorem 7.4). The framework rests on the Stratified Autopoietic Viability Geometric System (SAVGS, Section 3), a five-component object whose geometry is a stratified G_C-reduced principal bundle with $G_C \in \{O(r), SO(r), \mathrm{CO}(r)\}$ selected by the policy-fiber geometry; on this object, the viability-weighted curvature κ_V is well-defined, gauge-invariant, and falsifiable.

The viability-weighted curvature κ_V is the leading-order operational form of the cost functional C, defined on policy loops in the agent parameter space. It is constructed (Section 4) as the positive part of the directional derivative of a Bregman divergence h_α evaluated at the algorithmic rate-distortion distance $\mathrm{dist}_D(x)$, a single-string quantity that eliminates the deterministic-versus-random-coding gap of classical rate-distortion theory. For a circular loop γ_a of amplitude a in a radially symmetric viability landscape V with maximum $V_{\max}$ at the origin, the operational form reduces to

$$\kappa_V(a) \;=\; \frac{1}{V_{\max}}\,\langle V_{\max} - V \circ \gamma_a \rangle \;=\; a^2, \tag{1}$$

independent of dimension; the structure group $\mathrm{Stab}(\kappa_V)$ is then the endogenous structure group of the system and is computed, not chosen.

A Bregman-divergence Noether correspondence (Section 5, Proposition 5.1) supplies the affine-invariant structure required for the gauge invariance of κ_V: under a one-parameter affine transformation on the dual coordinate, the Bregman divergence is invariant, and Noether's theorem yields a conserved current. The correspondence is a theorem with explicit, checkable preconditions: the distortion measure and the source prior must both be Bregman divergences. A system that violates this precondition refutes the correspondence.

The principal contributions are summarized as follows.

1. The SAVGS framework (Definition 3.1), a five-component stratified object that resolves the type confusions, gauge ambiguities, and missing premises of prior formulations. The geometry is a stratified G_C-reduced principal bundle with $G_C \in \{O(r), SO(r), \mathrm{CO}(r)\}$ (Definition 2.2), and κ_V is computed on this object. The Fisher-minimal horizontal lift on each constant-active-set stratum is given in closed KKT form (Proposition 3.4); the projected differential inclusion at constraint-switching boundaries is given in Remark 3.5; the small-loop viability–holonomy theorem (Theorem 3.6) gives the leading-order ε^2 expansion of the endpoint margin and the sufficient condition for endpoint viability; the closure-aware curvature $\kappa_{\mathrm{closure}}$ of Definition 3.12 isolates autopoietic failure from behavioral hysteresis.

2. A single composition theorem (Theorem 7.4) for the seven optics in $\mathbf{Optic}(\mathcal{C})$, replacing the rhetorical "bridge-rung" construction of prior work. The composite is a well-defined typed endo-optic (Remark 7.8); under a realization functor and explicit contraction bound (Proposition 13.1), the realized update has a unique fixed point.

3. The algorithmic rate-distortion distance dist_D (Definition 4.1) and the derivation of κ_V from a Bregman divergence evaluated at dist_D (Proposition 4.4), supplying the deterministic single-string content required for the per-trajectory falsification protocol.

4. A Bregman-divergence Noether correspondence (Proposition 5.1) that turns the prior Noether analogy into a theorem with checkable preconditions.

5. A seven-claim falsification hierarchy (Definition 6.1) on the operational form κ_V, with five derivative claims (A–E) in the $n = 3$ prototype, an $n \geq 4$ non-abelian extension (Claim F), and a CPTP-Zeno quantum lift (Claim G).

6. Numerical confirmation of all seven claims within their stated tolerances, including heavy-tail index stress-testing of Claim D across the Student–t tail-parameter axis (Section 10) and generalization of A–E to the $n = 4$ non-abelian regime (Section 11).

7. An inverse-limit construction of the directed system of RAFs in **Optic**(**Set**), yielding the maximal RAF $R_{\max}$ as the filtered colimit (Construction 14.1).

8. A numerical Banach contraction argument for T over $X = [0,1]^d$ across a (d, k) robustness grid of 375 configurations, demonstrating that the unification object exists by Banach's theorem [6] throughout the extended setting (Section 13).

9. A main proposition (Proposition 15.1) stating the joint thesis in its strongest defensible form: on smooth constant-active- set strata, Fisher-minimal constraint-preserving adaptation defines a stratified connection whose viability-weighted curvature predicts leading-order policy hysteresis; whether that holonomy is fatal depends on viability margins, along-path disturbances, and regeneration of internal maintenance machinery.

The remainder of the article is organized as follows. Section 2 fixes the categorical and operational preliminaries. Section 3 constructs the SAVGS framework. Section 4 introduces the algorithmic rate-distortion distance and derives κ_V. Section 5 states the Bregman-Noether correspondence. Section 6 states the falsification hierarchy. Section 7 proves the composition theorem and lists the seven optics. Section 8 reports the foundational test results (Claims F and G). Section 9 operationalizes A–E in the $n = 3$ prototype. Section 10 stress-tests Claim D's heavy-tail index. Section 11 generalizes A–E to $n = 4$. Section 12 treats the CPTP-Zeno lift in detail (Claim G). Section 13 presents the T-iteration contraction results. Section 14 presents the filtered-colimit RAF construction. Section 15 states the main proposition. Section 16 discusses limitations, implications, and future directions. Section 17 concludes.

2 Preliminaries

Definition 2.1 (Viability depth functional D_V; distinct from curvature)**.** Let M be a smooth state manifold, $V : M \to \mathbb{R}$ a smooth viability function with maximum $V_{\max} = \max_M V$, and $\gamma_a : [0,1] \to M$ a smooth policy loop of amplitude a, parametrized at unit speed so that $|\gamma_a| = \int_0^1 |\dot\gamma_a(t)|\, dt$ is the loop length. The *viability depth functional* is

$$D_V(\gamma_a) \;=\; \frac{1}{|\gamma_a|}\int_{\gamma_a}\bigl(V_{\max} - V\bigr)\, ds \;=\; \frac{1}{|\gamma_a|}\int_0^1 \bigl(V_{\max} - V\circ\gamma_a(t)\bigr)\,|\dot\gamma_a(t)|\, dt. \tag{2}$$

For the radially symmetric viability $V(x_1, \ldots, x_{n-1}) = 1 - \sum_i x_i^2$ and the circular loop $\gamma_a(t) = (a\cos 2\pi t, a \sin 2\pi t, 0, \ldots, 0)$ at unit speed, $D_V(a) = a^2$. The viability depth is a loop-averaged deficit; it is *not* a curvature 2-form: it carries no orientation, no dependence on path ordering, and no commutator structure. The curvature-based viability-weighted curvature κ_V (Definition **??**, contracting the curvature 2-form F with viability covectors) is a distinct object; the equality $D_V(a) = \kappa_V(a) = a^2$ in the radial prototype is a model-specific relation, not an identity.

Definition 2.2 (Structure group: $O(r)$, $SO(r)$, or $CO(r)$ with Weyl scale)**.** The Fisher–Rao metric on a policy fiber of dimension r admits an orthonormal frame bundle with structure group $O(r)$ ([10, 3]). If the policy frame is oriented, the structure group reduces to $SO(r)$. A reduction to the conformal group $\mathrm{CO}(r) = \mathbb{R}_+ \ltimes \mathrm{SO}(r)$ requires an additional *scale (Weyl) structure* on the policy bundle: a positive scale bundle $L \to E$ together with a conformal class $[g^F]$ of the Fisher metric and a Weyl connection preserving the conformal class. The three cases are distinct modeling commitments; Chentsov's uniqueness theorem [2] fixes the Fisher metric under statistical-map invariance but does *not* by itself select among $O(r)$, $SO(r)$, or $\mathrm{CO}(r)$. For the manuscript's prototypes we use: $G_C = SO(2)$ for the abelian $n = 3$ prototype; $G_C = SO(3)$ for the non-abelian $n = 4$ prototype; and $\mathrm{CO}(r)$ only when an endogenous scale variable (effective precision, temperature, sample size, or Weyl factor) is explicitly modeled (Remark 2.3).

Remark 2.3 (Fisher–Weyl policy structure, when $\mathrm{CO}(r)$ is justified). A Fisher–Weyl policy structure consists of: (i) a policy bundle $E \to B$; (ii) a vertical Fisher metric g^F; (iii) a positive scale bundle $L \to E$; (iv) a conformal class $[g^F]$; (v) a Weyl connection preserving the conformal class. The conformal frame bundle then has structure group $\mathrm{CO}(r)$. If the cost functional depends only on the conformal class of g^F and is invariant under positive rescalings of orthonormal frames, its stabilizer contains $\mathrm{CO}(r)$; if scale is fixed, the stabilizer reduces to $SO(r)$ or $O(r)$. This preserves the $\mathrm{CO}(r)$ ambition in a mathematically honest way: $\mathrm{CO}(r)$ is an *added modeling structure*, not a consequence of Chentsov's theorem alone.

Remark 2.4 (Physical meaning of the $n = 3 \to n = 4$ transition). At $n = 3$ the policy simplex is two-dimensional, the Fisher–Rao stabilizer is $\mathrm{CO}(2) = \mathbb{R}_+ \ltimes \mathrm{SO}(2)$, and the Lie algebra $\mathfrak{so}(2)$ is one-dimensional, generated by the single infinitesimal rotation about the origin. Two rotations about the same origin necessarily commute (they lie in the same one-parameter subgroup), so the holonomy of a sequence of loops is path-independent up to homotopy, and the path-ordering in the parallel-transport exponential collapses to ordinary commutative multiplication. The non-abelian signature (Claim F) is then unobservable: the commutator $[R_1, R_2] = 0$ identically, regardless of the rotation amplitudes. At $n = 4$ the policy simplex is three-dimensional, the stabilizer is $\mathrm{CO}(3) = \mathbb{R}_+ \ltimes \mathrm{SO}(3)$, and $\mathfrak{so}(3)$ is three-dimensional with three independent generators $\{\mathfrak{l}_x, \mathfrak{l}_y, \mathfrak{l}_z\}$ satisfying $[\mathfrak{l}_z, \mathfrak{l}_x] = \mathfrak{l}_y$ etc. Two rotations about *distinct* axes (e.g. $R_z(\alpha_1)R_x(\alpha_2)$ vs $R_x(\alpha_2)R_z(\alpha_1)$) no longer commute, and the small-angle commutator is $[R_1, R_2] \approx \alpha_1\alpha_2[\mathfrak{l}_i, \mathfrak{l}_j]$ with Frobenius norm $\sqrt{2}\,\alpha_1\alpha_2$ (each $\mathfrak{so}(3)$ basis element has Frobenius norm $\sqrt{2}$). The non-abelian signature is therefore a quantitative, falsifiable prediction specific to $n \geq 4$; it cannot be tested at $n = 3$ by any experimental design.

Definition 2.5 (Optic category)**.** Following Riley [4], for a category $\mathcal{C}$ with finite limits the *optic category* $\mathbf{Optic}(\mathcal{C})$ has as objects triples (M, C, R) of a forward carrier M, a backward carrier C, and a residual R, and as morphisms pairs (φ, ρ) of a forward map $\varphi : M \to M'$ and a backward map $\rho : R' \to R$ satisfying the residual-compatibility condition. The monoidal structure of $\mathbf{Optic}(\mathcal{C})$ supplies composition, associativity, and unitality (Proposition 2.3 of [4]; see also [5]).

Definition 2.6 (RAF set)**.** Following Hordijk and Steel [7, 8], given a catalytic reaction network $(X, \mathcal{R}, F, c)$ over a molecule set X with food set $F \subset X$ and catalysis assignment $c : \mathcal{R} \to 2^X$, a subset $R' \subseteq \mathcal{R}$ is a *reflexively autocatalytic food-generated (RAF) set* if (i) every $r \in R'$ is catalyzed by some element of $F \cup \bigcup_{r' \in R'} \mathrm{react}(r')$, (ii) the food-generated closure of R' contains all reactants of R', and (iii) R' is closed under the induced closure operator.

Definition 2.7 (CPTP channel)**.** A *completely-positive trace-preserving (CPTP) channel* on a finite Hilbert space $\mathcal{H}$ is a linear map $\Phi : \mathrm{End}(\mathcal{H}) \to \mathrm{End}(\mathcal{H})$ of the form $\Phi(\rho) = \sum_i K_i\, \rho\, K_i^\dagger$ with Kraus operators $\{K_i\}$ satisfying $\sum_i K_i^\dagger K_i = I$. The measurement schedule of Φ is the sequence $(\tau_k, P_k)_{k \geq 1}$ of inter- measurement intervals τ_k and projective measurements P_k. The Liouvillian $\mathcal{L}$ of the underlying Markov process has a spectral gap $\Delta = \min_{\lambda \neq 0} \mathrm{Re}(\lambda)$.

Definition 2.8 (Bregman divergence)**.** A *Bregman divergence* [9, 3] generated by a strictly convex differentiable function ϕ is

$$D_\phi(p, q) \;=\; \phi(p) - \phi(q) - \langle \nabla\phi(q),\, p - q\rangle. \tag{3}$$

The divergence is not symmetric in general. The *dual affine coordinates* are p (the primal) and $\nabla\phi(p)$ (the dual). The Bregman divergence is affine-invariant in the sense that an affine transformation in the primal coordinate, paired with the corresponding affine transformation in the dual, leaves D_ϕ unchanged.

3 The SAVGS Framework

The Stratified Autopoietic Viability Geometric System (SAVGS) is the minimal object on which the viability-weighted curvature can be stated without the type confusions, gauge ambiguities, and missing premises of prior formulations. It assembles five components that prior work treats separately or treats at the wrong level of abstraction.

Definition 3.1 (SAVGS)**.** The SAVGS is the tuple $(B, E, \mathcal{P}, \varepsilon, \Gamma)$:

1. A continuous control base manifold $B \subset \mathbb{R}^d$ (denoted Θ in earlier drafts), with coordinates including food scarcity, danger intensity, and sensor noise.
2. A policy bundle $\pi : E \to B$ with total space E and policy fiber $\mathcal{P} = \pi^{-1}(\theta)$, where $\mathcal{P}$ is the open simplex $\Delta_\circ^{m-1}$ (for m actions), the circle S^1 (abelian $n = 3$ prototype), the rotation group $SO(3)$ (non-abelian $n = 4$ prototype), or another declared policy manifold. The bundle projection $\pi : E \to B$ is the canonical projection onto the base manifold (not the wrong-direction $\Delta^{n-1} \to \Theta$ of earlier drafts); on the trivial bundle $E = B \times \mathcal{P}$ this is $\pi(\theta, p) = \theta$.
3. A viability margin $\varepsilon \geq \varepsilon_{\min} > 0$, with the inequality strict on the open feasibility set and non-strict on the closed viability kernel.
4. An endogenous maintenance graph $\Gamma = (M, R, E_\Gamma)$ of maintenance operations M, regeneration rules R, and maintenance edges E_Γ (renamed to avoid clash with total space E).
5. A 2-categorical span $\mathrm{Stab}_1 \leftarrow \mathrm{Boundary} \to \mathrm{Stab}_2$ that resolves the constraint-switching boundary discontinuity between strata (Conjecture 16.1 for the full gluing theorem).

The five components compose into a single object whose geometry is a stratified G_C-reduced principal bundle over the control manifold, with policy fibers over each stratum and viability kernels defined by non-strict inequalities on each stratum. The viability-weighted curvature of Proposition 4.4 is computed on this object.

Remark 3.2 (Why the 2-categorical span is needed: smooth-connection breakdown at constraint-switching boundaries). The Ehresmann connection of the geometric construction requires smooth variation of horizontal subspaces across the base manifold. At a constraint-switching boundary—where the active set of inequality constraints on the viability margin ε changes—the horizontal subspace changes discontinuously. The connection is therefore no longer smooth at the boundary, and the standard holonomy integral breaks down. The fifth SAVGS component resolves this by introducing a 2-categorical span $\mathrm{Stab}_1 \leftarrow \mathrm{Boundary} \to \mathrm{Stab}_2$ that glues the two smooth strata via a boundary 2-cell; the connection is defined piecewise on each stratum and a matching condition on the boundary supplies the 2-categorical compatibility. The formal development of the 2-categorical structure (the lax-functorial gluing, the boundary 2-cell's coherence with the connection 1-forms, and the homotopy-theoretic well-definedness of the resulting holonomy) is left for future work; the operational requirement on the prototype of Sections 9–11 is that the active set remains constant along each loop, which places the loop entirely within a single smooth stratum and avoids the boundary discontinuity.

Remark 3.3 (Pathwise viability versus endpoint viability). A small loop with bounded endpoint holonomy does not imply that intermediate points of the loop are viable: the endpoint bound is a weaker statement than pathwise viability. To control intermediate points, the calibration protocol of Section 9.1 applies a Nagumo inward-pointing condition along the path, equivalently a viability tube of radius ε around the loop. A loop with bounded endpoint holonomy but intermediate viability violation is a false negative for the holonomy prediction; the tube check excludes it. The tube is operationalized by verifying that the entire ε-neighborhood of $\gamma_a([0, 1])$ lies inside the closed viability kernel $\{x : V(x) \geq V_{\max} - \varepsilon\}$; this is a binary check that produces a separate falsifiable verdict on the pathwise viability content of the holonomy prediction.

3.1 Rigorous Fisher-minimal transport law on constant-active-set strata

On a constant-active-set stratum of SAVGS, the active viability constraints are represented locally by $c(\theta, p) = 0$ with $J_p = D_p c$ of full row rank and $J_\theta = D_\theta c$. For an environmental velocity $\dot{\theta}$, the Fisher-minimal policy velocity is the unique solution of $J_p v + J_\theta \dot\theta = 0$ minimizing $\frac{1}{2} v^\top G(p) v$, where $G(p)$ is the Fisher information metric on the open simplex Δ^{n-1}.

Proposition 3.4 (Fisher-minimal horizontal lift (KKT form))**.** *On a constant-active-set stratum where J_p has full row rank, the unique Fisher-minimal policy velocity is*

$$v^*(\theta, p, \dot\theta) \; = \; -G(p)^{-1} J_p^\top \left(J_p \, G(p)^{-1} \, J_p^\top\right)^{-1} J_\theta \, \dot\theta, \tag{4}$$

which defines the connection 1-form $A_i = G^{-1} J_p^\top (J_p G^{-1} J_p^\top)^{-1} \partial_i c$ on the stratum, satisfying $\dot p = -A_i(\theta, p) \, \dot\theta^i$. The horizontal lift $H_i = \partial_{\theta^i} - A_i$ is a genuine smooth connection on each constant-rank stratum, with curvature $F_{ij} = \partial_i A_j - \partial_j A_i + [A_i, A_j]$.

Proof. The constrained minimum of $\frac{1}{2} v^\top G v$ subject to $J_p v + J_\theta \dot\theta = 0$ is found by the method of Lagrange multipliers. Setting $\nabla_v [\frac{1}{2} v^\top G v + \mu^\top (J_p v + J_\theta \dot\theta)] = G v + J_p^\top \mu = 0$ gives $v = -G^{-1} J_p^\top \mu$. Substituting into the constraint: $-J_p G^{-1} J_p^\top \mu + J_\theta \dot\theta = 0$, so $\mu = (J_p G^{-1} J_p^\top)^{-1} J_\theta \dot\theta$. The full-row-rank hypothesis on J_p guarantees the inverse exists. The horizontal subspace $H_{(\theta,p)} = \ker(J_p) \cap T_p \Delta^{n-1}$ varies smoothly in (θ, p) on the stratum (constant-rank hypothesis), so the resulting 1-form A_i is C^2; the curvature F_{ij} is then the standard Ehresmann-curvature expression by direct computation [10]. ☐

Remark 3.5 (Projected differential inclusion at constraint-switching boundaries). At a constraint-switching boundary—where an inequality enters or leaves the active set—the full-row-rank hypothesis of Proposition 3.4 fails: the rank of J_p jumps, and the smooth connection A_i is no longer defined. The correct replacement is the projected differential inclusion

$$\dot p \; \in \; \arg \min_{v \in T_{\mathcal{K}}(\theta, p; \dot\theta)} \; \tfrac{1}{2} \left\| v - v_0 \right\|_{F,p}^2, \tag{5}$$

where $T_{\mathcal{K}}(\theta, p; \dot\theta)$ is the Bouligand contingent cone to the viability kernel $\mathcal{K}$ at (θ, p) in the direction $\dot\theta$ and v_0 is the unconstrained Fisher-minimal velocity. This is viability dynamics, not ordinary smooth-connection theory; the SAVGS 2-categorical span of Definition 3.1 glues the smooth-stratum connections across the boundary via the matching condition on the boundary 2-cell.

Theorem 3.6 (Small-loop viability–holonomy)**.** *Let $(\Theta, \pi, \Delta^{n-1}, \varepsilon, \Gamma)$ be a SAVGS as in Definition 3.1, on a constant-active-set stratum where Proposition 3.4 applies and the connection 1-form A_i is C^2. Let $\gamma_\varepsilon : [0, 1] \to \Theta$ be a square loop of side ε in the (θ^1, θ^2)-plane at θ_0, contained entirely in a single stratum, and let $h_\alpha(\theta, x)$ be a C^2 viability function with $h_\alpha(\theta_0, x_0) > 0$. Then the parallel transport of the policy around γ_ε satisfies*

$$p_{\text{end}} - p_0 \; = \; \varepsilon^2 \, F_{12}(\theta_0, p_0) \; + \; O(\varepsilon^3), \tag{6}$$

up to the orientation convention for F_{12}. The induced change in the viability margin is

$$h_\alpha(x_{\text{end}}) - h_\alpha(x_0) \; = \; \varepsilon^2 \, D_p h_\alpha\big(F_{12}(\theta_0, p_0)\big) \; + \; O(\varepsilon^3) \; + \; \Delta_\alpha^{\text{other}}, \tag{7}$$

where $\Delta_\alpha^{\text{other}}$ collects all non-returning energy, integrity, repair, and environmental contributions along the loop. In particular, the endpoint remains viable whenever

$$h_\alpha(x_0) \; + \; \varepsilon^2 \, D_p h_\alpha(F_{12}) \; - \; C_\alpha \, \varepsilon^3 \; - \; |\Delta_\alpha^{\text{other}}| \; > \; 0 \qquad \textit{for every } \alpha, \tag{8}$$

where C_α depends on the C^2 bounds of h_α and A_i.

Proof. The expansion (6) is the standard non-abelian Stokes formula for the parallel transport around an infinitesimal loop (see, e.g., [10, 11]): writing the path-ordered exponential $\mathcal{P} \exp(-\oint A) = 1 - \iint_S F_{12} \, d\theta^1 d\theta^2 + O(\varepsilon^3)$ on a square S of side ε gives $p_{\text{end}} - p_0 = -\varepsilon^2 F_{12} + O(\varepsilon^3)$ (the sign convention is fixed by the orientation of A_i and is absorbed into F_{12}). Equation (7) follows by Taylor expansion of h_α at x_0 to first order in $p_{\text{end}} - p_0$ and second order in ε, with $\Delta_\alpha^{\text{other}}$ collecting the integral of $D_\theta h_\alpha \, \dot\theta$ along the loop plus the $O(\varepsilon^3)$ remainder. The sufficient condition (8) is then the requirement that the worst-case $O(\varepsilon^3)$ remainder plus the non-returning contribution not exceed the initial margin minus the leading-order curvature contribution. ☐

Remark 3.7 (What Theorem 3.6 does and does not claim). Theorem 3.6 bounds the *endpoint* viability margin by the leading-order curvature contribution plus an $O(\varepsilon^3)$ remainder. It does *not* bound the *pathwise* viability margin: a loop with bounded endpoint holonomy may transit non-viable regions. Pathwise viability requires the additional Nagumo inward-pointing condition of Remark 3.3—equivalently a viability tube of radius ε around $\gamma_a([0,1])$ contained in the closed viability kernel. Curvature is therefore neither necessary nor sufficient for survival in isolation; the bidirectional "equivalence" is replaced by the unidirectional upper bound "vulnerability $\leq \kappa_V$" of Proposition 15.1, with the trivial lower bound 0 attained when $\kappa_V \equiv 0$.

3.2 Square-root embedding and gauge invariance

Proposition 3.8 (Square-root embedding)**.** *The square-root embedding $\psi_a = 2\sqrt{p_a}$ places the open simplex Δ^{n-1} on the positive orthant of the unit sphere $S^{n-1} \subset \mathbb{R}^n$. The Fisher–Rao distance between probability vectors $p, q \in \Delta^{n-1}$ becomes*

$$d_{\mathrm{FR}}(p,q) \;=\; 2\,\arccos\Big(\sum_a \sqrt{p_a q_a}\Big), \tag{9}$$

the standard round metric on the positive orthant. All geometric quantities computed in this embedding are gauge-invariant; the logit coordinates of prior formulations are recovered as a stereographic projection.

Proof. Direct computation of the Fisher–Rao metric tensor in the square-root embedding [12, 3]. The induced metric on the positive orthant of S^{n-1} is the standard round metric; the stereographic projection onto the tangent plane at any fixed reference point gives the logit coordinates. The gauge-invariance follows from the round metric's $\mathrm{SO}(n)$-invariance. □

3.3 Intervention-based autopoiesis closure test

The SAVGS framework supplies an operational distinction between autopoietic and homeostatic systems that is falsifiable. The audit's core distinction is between *homeostasis* (the agent remains inside a prescribed viable set, possibly via externally supplied maintenance) and *autopoiesis* (the agent regenerates the sensing, control, repair, and boundary-maintenance machinery required to remain there). The maintenance graph $\Gamma = (M, R, E)$ of Definition 3.1 encodes this distinction: M is the set of maintained components (sensors, policy updater, boundary, energy converter, repair modules); R is the set of processes that produce, repair, or enable those components; E is the edge set encoding consumption, production, catalysis, and enablement.

Definition 3.9 (Operational closure criteria)**.** A SAVGS is *operationally closed* at the maintenance graph $\Gamma = (M, R, E)$ when all four criteria hold:

1. Every indispensable component $m \in M_{\mathrm{ess}} \subseteq M$ degrades at a strictly positive rate.
2. Every $m \in M_{\mathrm{ess}}$ is regenerated by at least one internal process $r \in R$.
3. Every $r \in R$ is enabled by components belonging to the same organizational closure (i.e., the enablers of r lie in M_{ess}).
4. External inputs may supply matter and free energy, but not preassembled replacements for any $m \in M_{\mathrm{ess}}$.

The indispensable maintenance subgraph $\Gamma_{\mathrm{ess}} \subseteq \Gamma$ is required to lie in a productive strongly connected component, and the associated fluxes must support a feasible positive steady state (a flux feasibility linear program on Γ must admit a strictly positive solution).

Definition 3.10 (Autopoiesis closure test)**.** For each purportedly self-maintained component $m_j \in M_{\mathrm{ess}}$: (i) set its internal repair flux to zero; (ii) keep external energy and raw-material supply unchanged; (iii) apply the regeneration rules R for a fixed number of steps T; (iv) observe whether m_j reappears in the regenerated graph; (v) restore the repair pathway and test recovery. A component m_j is *causally internal* to the organizational closure iff its intervention produces the predicted downstream effect (loss of m_j causes loss of components $m_{j'}$ whose enabling chain depends on m_j) and the restoration test produces recovery within T steps. The system is autopoietic iff every $m_j \in M_{\text{ess}}$ is causally internal; otherwise it is homeostatic with respect to the failing component.

Remark 3.11 (Why the test is non-circular). The closure test of Definition 3.10 prevents "autopoiesis" from being satisfied by a simulator that silently repairs the agent from outside: the external intervention in step (i) is the only externally imposed change; the recovery in step (v) must be produced by the endogenous regeneration rules R alone. A homeostatic system that depends on a hidden external repair mechanism fails the test on step (v): the removed component does not reappear within T steps. The test is falsifiable: it predicts that removing a maintenance node from an autopoietic system causes the node (and its downstream dependents) to reappear within T steps; a system for which this prediction fails for any $m_j \in M_{\text{ess}}$ is classified homeostatic, not autopoietic.

Definition 3.12 (Closure-aware viability curvature)**.** Restricting the viability covectors of κ_V (Proposition 4.4) to the repair-machinery margins $h_j^{\text{repair}} = r_j - r_{j,\min}$ gives the *closure-aware viability curvature*

$$\kappa_{\text{closure}}(\theta, x) \;=\; \sup_{\|u\wedge v\|_{g_F}=1} \; \max_{j\in M_{\text{ess}}} \frac{\left[-Dh_j^{\text{repair}}(F(u,v))\right]^+}{h_j^{\text{repair}}(\theta,x)}. \tag{10}$$

The quantity κ_{closure} distinguishes three failure modes: (i) harmless behavioral hysteresis (the curvature projects onto non-maintenance directions); (ii) metabolic viability erosion ($\kappa_V > 0$ but $\kappa_{\text{closure}} = 0$); (iii) autopoietic failure proper ($\kappa_{\text{closure}} > 0$, the curvature erodes the machinery that makes future adaptation possible). Only the third is specifically autopoietic failure.

4 Algorithmic Rate-Distortion and Derivation of κ_V

The curvature-based viability-weighted curvature κ_V of Proposition 4.4 is the operational form of a directional derivative of a Bregman divergence evaluated at the algorithmic rate-distortion distance. The viability depth functional D_V of Definition 2.1 is a distinct object (loop-averaged deficit, not curvature). The algorithmic rate-distortion distance is a single-string quantity, defined per input x, and is intrinsically deterministic. It eliminates the type confusion between the set-average rate $R(D)$ of classical rate-distortion theory [13] and the single-string Kolmogorov complexity $K(x)$.

Definition 4.1 (Algorithmic rate-distortion distance)**.** Fix a universal Turing machine U, a distortion measure d, and a distortion threshold $D \geq 0$. The *algorithmic rate-distortion distance* of x at threshold D is

$$\text{dist}_D(x) \;=\; \min\big\{|p| : U(p) \text{ outputs } \hat{x},\; d(x,\hat{x}) \leq D\big\}. \tag{11}$$

Proposition 4.2 (Properties of dist_D)**.** *The function* dist_D *satisfies: (i) monotone non-increasing in D; (ii) bounded above by $K(x)$ (on setting D to the trivial distortion that accepts any output); (iii) bounded below by* 0*; (iv) upper- semicomputable by dovetailing over all programs.*

Proof. (i) monotonicity by inspection of the definition: larger D admits more candidate programs, hence no larger a minimum. (ii) The trivial distortion accepts any output, so the empty program qualifies and $\text{dist}_D(x) \leq K(x)$. (iii) Trivial. (iv) Standard dovetailing: enumerate programs in order of length, simulate each dovetailed step, halt each when its output $\hat{x}$ satisfies $d(x,\hat{x}) \leq D$; the minimum length is approached from above. □

Remark 4.3 (No deterministic-versus-random-coding gap). The gap $R_{\mathrm{det}}(D) \geq R(D)$ of classical rate-distortion theory arises because $R(D)$ is a set-average quantity whose achievability requires random coding. $\mathrm{dist}_D(x)$ is a single-string quantity whose achievability is automatic: the program p that achieves the minimum exists by definition. The gap does not arise.

Proposition 4.4 (Derivation of κ_V)**.** *Let $h_\alpha(\theta, x) = D_\phi\big(\mathrm{dist}_D(x), \mathrm{dist}_D(x_0)\big)$ be a Bregman divergence evaluated at the algorithmic rate-distortion distance of the current state x relative to a reference x_0. Let $F(u, v)$ be the curvature 2-form of the G_C-connection on* SAVGS*. The per-pair viability-weighted curvature is*

$$\kappa_\alpha(\theta, x; u, v) \;=\; \frac{\big[-D_{F(u,v)} h_\alpha\big]^+(\theta, x)}{h_\alpha(\theta, x)}, \tag{12}$$

where $[\,\cdot\,]^+$ denotes the positive part. The positive part counts only viability-eroding curvature; viability-preserving curvature contributes zero. The aggregate index is the worst case over unit-area bivectors and over the active viability covectors:

$$\kappa_V(\theta, x) \;=\; \sup_{\substack{u,v \in T_\theta\Theta \\ \|u \wedge v\|_{g_F} = 1}} \; \max_{\alpha : h_\alpha(\theta, x) > 0} \; \kappa_\alpha(\theta, x; u, v). \tag{13}$$

The quantity $\kappa_V(\theta, x)$ is the worst fractional loss of viability margin per unit oriented environmental-loop area, to leading order; it is more meaningful than $\|F\|$ alone because it contracts the geometric defect with the experimentally specified survival covectors Dh_α.

Proof. Direct construction: the algorithmic rate-distortion distance $\mathrm{dist}_D(x)$ of Definition 4.1 supplies the function h_α through the Bregman divergence of Definition 2.8; the Bregman divergence supplies the affine-invariant structure required for the G-invariance of the Noether correspondence (Section 5); the positive part of the directional derivative counts only viability-eroding contributions. The aggregate form (13) contracts the geometric 2-form F with the survival covectors Dh_α in the worst case over unit bivectors (so κ_V is invariant to the choice of (u, v) used to probe the curvature) and over active viability margins (so κ_V detects the most vulnerable margin at the point). Gauge invariance follows from Proposition 3.8 (square-root embedding) and Proposition 5.1 (Bregman Noether); the wedge-norm constraint $\|u \wedge v\|_{g_F} = 1$ makes κ_V dimensionally a per-area quantity, matching the leading-order expansion (7) of Theorem 3.6. $\square$

Remark 4.5. The resulting κ_V is an observable quantity computed from a single trajectory of the system, not a set-average over an ensemble. This matches the deterministic structure of the RAF transition (Definition 2.6) and supplies the per-trajectory operational form required by the falsification protocol of Section 6.

4.1 Smooth finite-code surrogate and algorithmic upper envelope

The algorithmic rate-distortion distance dist_D of Definition 4.1 is upper-semicomputable but generally uncomputable, integer-valued, and not differentiable. It cannot serve as the argument of a directional derivative in any classical sense. To obtain an operational, differentiable object suitable for Proposition 4.4, we replace dist_D with a *smooth finite-code surrogate*.

Definition 4.6 (Smooth finite-code rate-distortion surrogate)**.** Let $\{c_j\}_{j=1}^N$ be a finite prefix-free code family with code lengths $\ell(c_j)$ and reconstructions $\hat{x}_j = \mathrm{dec}(c_j)$. For a distortion measure d, threshold $D \geq 0$, and parameters $\tau > 0$, $\beta > 0$, define

$$r_{\tau,\beta,D}(x) \;=\; -\tau \, \log \sum_{j=1}^{N} 2^{-\ell(c_j)/\tau} \, \exp\!\Big(-\beta \, [d(x, \hat{x}_j) - D]_+^2 / \tau\Big). \tag{14}$$

If $x \mapsto d(x, \hat{x}_j)$ is C^2 for each j, then $r_{\tau,\beta,D} \in C^2(\mathbb{R}^n, \mathbb{R})$, as the sum, product, and composition of smooth maps are smooth, and the exponential damping controls the tail behavior.

Proposition 4.7 (Operational viability margin from smooth surrogate)**.** *Replace* $\mathrm{dist}_D(x)$ *in the construction of Proposition 4.4 with the smooth surrogate* $r_{\tau,\beta,D}(x)$*. Define the viability margin* $h_\alpha(\theta, x) = D_\phi(r_{\tau,\beta,D}(x),\, r_{\tau,\beta,D}(x_0))$*. Then the directional derivative* $D_{F(u,v)}h_\alpha$ *is well-defined, the viability-weighted curvature* $\kappa_V(\theta, x)$ *of Proposition 4.4 is a smooth observable quantity, and the small-loop Theorem 3.6 applies with* h_α *in place of the classical* dist_D*-based form.*

Proof. The smoothness hypothesis on $x \mapsto d(x, \hat{x}_j)$ gives $r_{\tau,\beta,D} \in C^2$; the Bregman divergence D_ϕ of a C^2 strictly convex generator is C^2 in its arguments (Definition 2.8); the composition $h_\alpha = D_\phi \circ r_{\tau,\beta,D}$ is therefore C^2, and the directional derivative $D_{F(u,v)}h_\alpha$ is well-defined as a classical derivative. All subsequent constructions (positive-part projection, sup over unit bivectors, max over α) preserve smoothness, giving a C^2 observable κ_V. The Theorem 3.6 expansion is then a Taylor expansion of a C^2 function, valid by the standard C^2 hypothesis. □

Conjecture 4.8 (Algorithmic upper envelope)**.** There exists an upper-semicomputable algorithmic viability curvature κ_V^{alg} based on the algorithmic rate-distortion distance dist_D (Definition 4.1) such that, for any smooth finite-code surrogate $r_{\tau,\beta,D}$ of Definition 4.6,

$$\kappa_V^{\mathrm{surrogate}}(\theta, x) \;\le\; \kappa_V^{\mathrm{alg}}(\theta, x) \;+\; O(1),$$

in an appropriate asymptotic regime where the code family $\{c_j\}$ expands to cover all programs of bounded length. This conjecture preserves the algorithmic-rate-distortion ambition without falsely treating dist_D as a differentiable quantity.

Corollary 4.9 (Geometric-phase form of κ_V for radially symmetric V)**.** *Let* $V : M \to \mathbb{R}$ *be a smooth viability function with isolated maximum* $V_{\max}$ *at the origin, whose level sets* $\{V = V_{\max} - r\}$ *are spheres of radius* r *for* $r \in [0, r_\star]$ *(i.e.,* V *is radially symmetric on a neighborhood of* $V_{\max}$*). Let* $\gamma_a : [0,1] \to M$ *be a circular loop of amplitude* $a < r_\star$ *in any* 2*-plane through the origin. Then*

$$\kappa_V(\gamma_a) \;=\; \frac{1}{V_{\max}}\,\langle V_{\max} - V \circ \gamma_a\rangle_{[0,1]} \;=\; \frac{1}{V_{\max}}\,\overline{\delta V}(a), \tag{15}$$

where $\overline{\delta V}(a)$ *is the loop-averaged radial deficit at radius* a*. In particular,* κ_V *is a function of* a *alone (independent of the loop's plane through the origin), and the geometric holonomy* $H_{\mathrm{geo}}(\gamma_a) = \pi\,\kappa_V(\gamma_a)$ *equals the area* πa^2 *in the quadratic-deficit case* $\overline{\delta V}(a) = a^2$*, matching the* $n = 3$ *operational form of Section 9–11.*

Proof. By radial symmetry $V \circ \gamma_a(t)$ depends only on a (the loop radius), not on t (the loop phase) or on the choice of 2-plane through the origin. Hence $\langle V_{\max} - V \circ \gamma_a\rangle = V_{\max} - V \circ \gamma_a(0) = \overline{\delta V}(a)$. The geometric holonomy equals the loop area times the (constant) curvature 2-form on a radially symmetric landscape, by Stokes $\oint_\gamma A = \iint_D F$; substituting $F = \kappa_V$ and $\mathrm{area}(\gamma_a) = \pi a^2$ gives $H_{\mathrm{geo}} = \pi a^2$ in the quadratic-deficit case. The dimension-independence (the same formula holds at $n = 3$ and $n = 4$) follows from the radial symmetry holding in any 2-plane through the origin of $\mathbb{R}^{n-1}$. □

Remark 4.10 (MDP versus POMDP base space). The SAVGS base space $\Theta \subset \mathbb{R}^d$ is a continuous experimental control manifold, not the discrete grid-state space of a Markov decision process. When the underlying agent operates on a fully observable grid, the correct decision-theoretic frame is an MDP and Θ should be parameterized by environmental knobs (food scarcity, hazard intensity, sensor noise, energy price, repair latency) rather than by grid location. When the agent has only partial or noisy state access, the frame is a POMDP and the belief state takes the place of the grid state in the policy fiber. In both cases, the geometric loop lives in Θ, not in grid-state space; the agent acts inside the grid-world at each fixed θ, and infinitesimal loops in Θ are well-defined because Θ is a smooth manifold. The differential-curvature construction of κ_V (Proposition 4.4) is therefore well-defined on Θ regardless of the MDP/POMDP distinction; the latter only determines the structure of the policy fiber $\pi^{-1}(\theta)$.

Remark 4.11 (Counterexamples to curvature-survival equivalence). The bidirectional "curvature-survival equivalence" is false in general, as the audit observes: (i) a flat connection ($F \equiv 0$, hence $\kappa_V \equiv 0$) can transport the system arbitrarily far from $V_{\max}$ along a radial policy direction, exhausting the viability margin without any curvature contribution; (ii) a curved connection ($\kappa_V > 0$) on a large viability region ($V_{\max} \gg \kappa_V$) keeps the system viable, since the fractional loss $\kappa_V / V_{\max}$ is small. The correct statement is therefore *unidirectional*: κ_V upper-bounds the worst fractional loss of viability margin per unit loop area, but viability also depends on along-path drift, resource costs, and the regeneration of maintenance machinery (Propositions 15.1, 15.2). The "equivalence" is replaced by the upper-bound form "vulnerability $\leq \kappa_V$"; the lower bound 0 is trivially attained when $\kappa_V \equiv 0$.

5 Bregman-Divergence Noether Correspondence

A Noether-type correspondence for the Lagrangian

$$L \;=\; \mathbb{E}[d] + \lambda\, I \tag{16}$$

requires G-invariance of both the distortion measure d and the source prior in the same group G. Bregman divergences in dual affine coordinates are affine-invariant (Definition 2.8), which supplies both.

Proposition 5.1 (Bregman–Hessian Noether correspondence). *Let $Q \subset \mathbb{R}^r$ be open convex, $\phi \in C^3$ strictly convex, and $g_\phi = \nabla^2\phi$ the Hessian metric on Q. Let $L(q,\dot q) = \frac{1}{2} g_\phi(q)(\dot q, \dot q) - U(q)$ be the geodesic Lagrangian with potential $U \in C^2$, and let $S[q] = \int_a^b L(q,\dot q)\,dt$ be the action on the configuration space of smooth curves $q : [a,b] \to Q$. Let ξ be a complete affine vector field on Q whose flow $g_t = \mathrm{flow}_t(\xi)$ preserves the Hessian metric,*

$$\mathcal{L}_\xi\, g_\phi \;=\; 0, \tag{17}$$

and preserves the potential, $\xi(U) = 0$. Then S is invariant under the lifted flow $\tilde g_t \cdot (q, \dot q) = (g_t \cdot q,\; Dg_t \cdot \dot q)$, and by Noether's theorem [14, 15] the momentum

$$J_\xi(q,\dot q) \;=\; g_\phi(q)(\dot q, \xi(q)) \;=\; \langle \nabla\phi(q), \xi(q)\rangle \cdot \dot q_{\mathrm{dual}} \tag{18}$$

is conserved along Euler–Lagrange trajectories of S: $\frac{d}{dt} J_\xi(q(t), \dot q(t)) = 0$ on-shell.

Proof. The Hessian-metric preservation (17) gives $\mathcal{L}_\xi L = \xi(U) = 0$ (the kinetic term is invariant by $\mathcal{L}_\xi g_\phi = 0$, and the potential term is invariant by $\xi(U) = 0$); hence $S[\tilde g_t \cdot q] = S[q]$ for all t, and $\tilde g_t$ is a continuous variational symmetry of S. By Noether's theorem in its standard field-theoretic form [15], the one-parameter symmetry yields the conserved momentum $J_\xi(q,\dot q) = \partial L/\partial \dot q \cdot \xi(q) = g_\phi(q)(\dot q, \xi(q))$. The dual-coordinate form follows by identifying $\partial L/\partial \dot q = \nabla\phi(q) \cdot \dot q_{\mathrm{dual}}$ via the dual-coordinate structure of Bregman divergences (Definition 2.8). □

Corollary 5.2 (Affine Bregman invariance; sufficient condition). *A one-parameter affine flow g_t on Q that satisfies $\phi \circ g_t = \phi + \ell_t$ for an affine function ℓ_t preserves the Hessian metric $g_\phi = \nabla^2\phi$ (since $\nabla^2(\phi \circ g_t) = \nabla^2\phi$ on the affine transformation g_t by the chain rule, and $\nabla^2 \ell_t = 0$). Therefore the hypothesis (17) is satisfied, and the Bregman divergence D_ϕ is invariant in the paired primal-dual sense required by Proposition 5.1.*

Proposition 5.3 (Falsifiable precondition check). *The Bregman–Hessian Noether correspondence has a falsifiable precondition check: verify (i) that ϕ is C^3 strictly convex (positive-definite Hessian $g_\phi = \nabla^2\phi \succ 0$), (ii) that the vector field ξ generates an affine Hessian isometry ($\mathcal{L}_\xi g_\phi = 0$, equivalently $\mathcal{L}_\xi(\nabla^2\phi) = 0$), and (iii) that $\xi(U) = 0$. If all three conditions hold, the correspondence applies; failure of any condition refutes the theorem at that point.*

Proof. Operational verification: (i) compute the Hessian $\nabla^2\phi$ and verify positive definiteness; (ii) compute the Lie derivative $\mathcal{L}_\xi(\nabla^2\phi)$ via Cartan's formula $\mathcal{L}_\xi T = [i_\xi, d]T$ and verify it vanishes; (iii) compute the directional derivative $\xi(U)$ and verify it is zero. All three checks are mechanical and produce a binary verdict. □

Remark 5.4 (Application to gauge invariance of κ_V). If the policy connection and viability margins of Proposition 4.4 are constructed from a Bregman generator ϕ and the gauge group acts by affine Hessian isometries preserving the margins, the curvature F and the viability-weighted curvature κ_V are gauge-covariant (and gauge-invariant when the margins themselves are gauge-invariant). This is the rigorous replacement for the earlier Bregman-Noether proposition: not divergence invariance alone (which is false in general under arbitrary affine transformations holding ϕ fixed), but affine isometries of the Hessian metric yield conserved momenta by Noether's theorem.

6 The Seven-Claim Falsification Hierarchy

Definition 6.1 (Falsification hierarchy)**.** The viability-weighted curvature κ_V of Definition 2.1 (viability depth) and Proposition 4.4 (curvature-based κ_V) generate a seven-claim falsification hierarchy on the prototype of Section 9. Each claim is a specific quantitative prediction; failure of any prediction refutes the corresponding level of the hierarchy.

A. (Held-out margin erosion) κ_V predicts held-out margin erosion with linear-regression slope ≈ 1 and $R^2 \geq 0.9$.

B. (Orientation reversal amplitude) The holonomy $H(a) = \pi a^2$ reverses orientation when $|H| > \pi$, i.e. at $a_{\mathrm{rev}} = 1$.

C. (Holonomy-area scaling) $H(a) = c_1 a^2 + c_2 a^{3/2}$ with $c_1 \sim \pi$, $c_2 \sim C_{\mathrm{fat}} \neq 0$, $R^2 \geq 0.95$.

D. (Repeated-loop fatigue) Two-sided bound: the *conservative sufficient safety condition* $\sum_{k=1}^{K}(a_k\,(\kappa_V)_k + C_{\mathrm{fat}}\,a_k^{3/2} + \eta_k) < 1$ guarantees survival through K loops, and the *failure condition* $\sum_{k=1}^{K}(a_k\,\kappa_V(a_k) + C_{\mathrm{fat}}\,a_k^{3/2} + \eta_k) > 1$ predicts fatigue failure; $V_{\max,K} = \prod_k(1-F_k) < e^{-1}$ at K_{obs}; relative error $< 15\%$.

E. (Total-variance discrimination) $T = |H_{\mathrm{corr}} - H_{\mathrm{geo}}|/\sigma_{\mathrm{total}}$ is small under the loop condition and large under the no-loop control, with $T_{\mathrm{control}}/T_{\mathrm{loop}} \geq 5$.

F. (Non-abelian commuting control) At $n \geq 4$, the structure group G_C has non-abelian $\mathfrak{so}(3)$ when $G_C = SO(3)$; same-axis rotations commute (machine precision), distinct-plane rotations do not (nonzero non-abelian signature).

G. (CPTP-Zeno lift) The CPTP lift of the self- referential predictive paradox resolves via the Zeno measurement schedule, with Zeno scaling exponent $\alpha_{\mathrm{Zeno}} \approx 2$ distinguishable from the classical $\alpha_{\mathrm{cl}} \approx 1$.

The hierarchy is cumulative: failure at level k renders the subsequent levels operationally meaningless but does not refute the underlying categorical construction (Theorem 7.4).

Remark 6.2 (Recommended experimental ordering). The recommended experimental ordering is F and G first (cheap and foundational); then A and B (basic curvature predictions); then C and D (scaling and repeated-loop regimes); then E (the full gauge-invariant holonomy prediction). This ordering implements the principle that cheap foundational tests should precede expensive derivative tests. If F or G fails, the derivative tests are unnecessary.

7 The Single Composition Theorem

Construction 7.1 (Seven-optic endofunctor)**.** Let $\mathcal{C}$ be a category with finite limits. Let $\mathcal{O}_1, \ldots, \mathcal{O}_7 \in \mathbf{Optic}(\mathcal{C})$ be the seven optics corresponding to the seven arcs of domain unification:

1. $\mathcal{O}_1$: the RAF optic (forward = catalytic closure, backward = decoder, residual = the viability-kernel distance $D_\varphi(\mathrm{dist}_D(R), \mathrm{dist}_D(R_0))$);
2. $\mathcal{O}_2$: the RPSI optic (forward = CPTP predictive update, backward = posterior decoder, residual = Holevo information $I(\rho_{\mathrm{out}}; \hat{\rho}_{\mathrm{in}})$);
3. $\mathcal{O}_3$: the IFS optic (forward = iterated-function-system contraction [16, 17], backward = decoder, residual = contraction quotient c_i of f_i);
4. $\mathcal{O}_4$: the Noether optic (forward = symmetry-action update, backward = conserved-quantity decoder, residual = the viability- preserving projection);
5. $\mathcal{O}_5$: the perturbation optic (forward = perturbed Blahut–Arimoto update [18, 19], backward = perturbation decoder, residual = perturbation tensor $\partial d/\partial p$);
6. $\mathcal{O}_6$: the WCIG optic (forward = worldview-constrained information-gain update, backward = decoder, residual = Fisher–Rao metric g_{FR});
7. $\mathcal{O}_7$: the $n = 3$ Fisher–Rao optic (forward = parallel- transport update on SO(2), backward = decoder, residual = loop-area invariant; for $n \geq 4$, lifted to SO$(n-1)$ with viability margin ε and maintenance graph Γ).

The compatibility condition $A_i = M_{i+1}$ holds by construction.

Remark 7.2 (Optic decompositions of the arcs). The forward components are the encoding operations of each arc; the backward components are the decoding operations; the residuals are the information that flows from the forward pass to the backward pass. The IFS optic ($\mathcal{O}_3$) is a pure coalgebra [20] on the category of compact metric spaces, while the perturbation optic ($\mathcal{O}_5$) is a coalgebra with residual, the prototypical BA structure. The compatibility condition is satisfied by construction: each arc's forward action is the next arc's backward state. Two arcs deserve comment. The RPSI arc's forward component is a CPTP channel, not a deterministic function, because the RPSI self-reference paradox requires the quantum lift of Section 12. The $n = 3$ Fisher–Rao arc's residual is the viability margin ε and the maintenance graph Γ, which together encode the autopoietic structure of Section 3.

Table 1: Seven-optic composition. Each arc is decomposed as an optic (M_i, C_i, R_i) in $\mathbf{Optic}(\mathcal{C})$ with forward φ_i (encoder), backward ρ_i (decoder), and residual Res_i (information flowing forward to backward). The compatibility condition $A_i = M_{i+1}$ holds by construction. The total residual of T is the product $\mathrm{Res}_1 \times \cdots \times \mathrm{Res}_7$ in $\mathcal{C}$.

Arc	Forward φ_i (encoder)	Backward ρ_i (decoder)	Residual Res_i
$\mathcal{O}_1$ RAF	catalytic closure	RAF transition decoder	$D_\varphi(\mathrm{dist}_D(R), \mathrm{dist}_D(R_0))$
$\mathcal{O}_2$ RPSI	CPTP predictive update	posterior decoder	$I(\rho_{\mathrm{out}}; \hat{\rho}_{\mathrm{in}})$ (Holevo)
$\mathcal{O}_3$ IFS	Hutchinson contraction	deconvolution	contraction quotient c_i
$\mathcal{O}_4$ Noether	symmetry-action update	conserved-quantity decoder	viability-preserving projection
$\mathcal{O}_5$ Perturb.	perturbed BA update	perturbation decoder	perturbation tensor $\partial d/\partial p$
$\mathcal{O}_6$ WCIG	worldview-constrained IG update	decoder	Fisher–Rao metric g_{FR}
$\mathcal{O}_7$ n=3 FR	parallel transport on SO(2)	decoder	loop-area invariant (ε, Γ)

Remark 7.3 (Physical reading of the seven-optic composition). The seven-fold composition admits a single-sentence physical reading: a self-maintaining agent (RAF, $\mathcal{O}_1$) computes its next internal state by predicting its own sensory input (RPSI, $\mathcal{O}_2$), which under the Zeno lift is a quantum channel; the prediction contracts toward a stable perceptual fixed point (IFS, $\mathcal{O}_3$); the conserved quantity under the agent's symmetry group stabilizes the contraction (Noether, $\mathcal{O}_4$); small perturbations of the encoding cost are absorbed by the Blahut–Arimoto update (perturbation, $\mathcal{O}_5$); the agent's worldview constrains information gain (WCIG, $\mathcal{O}_6$); and the policy is finally transported on the Fisher–Rao sphere of probability parameters ($n = 3$ Fisher–Rao, $\mathcal{O}_7$).

The compatibility condition $A_i = M_{i+1}$ expresses that the encoding output of arc i is the state on which arc $i+1$ acts. The fixed point of T is then the self-consistent "agent-state-prediction-policy" quadruple, whose existence is decoupled from definitional rhetoric and reduced to the empirical contraction-rate question settled in Section 13.

Theorem 7.4 (Composition)**.** *Under Construction 7.1, the seven-fold composition*

$$T \;=\; \mathcal{O}_7 \circ \mathcal{O}_6 \circ \cdots \circ \mathcal{O}_1 \tag{19}$$

is a well-defined endofunctor on $\mathbf{Optic}(\mathcal{C})$*. The composition is associative and unital. The residual of* T *is the product* $\mathrm{Res}_1 \times \cdots \times \mathrm{Res}_7$ *in* $\mathcal{C}$*. The fixed point of* T*, when it exists, is the* unification object*.*

Proof. $\mathbf{Optic}(\mathcal{C})$ is a monoidal category under optic composition: the tensor product is $\mathcal{O}_2 \circ \mathcal{O}_1$ with the identity optic as unit; the associativity and unitality axioms follow from the universal property of the product $\mathrm{Res}_1 \times \mathrm{Res}_2$ in $\mathcal{C}$ (Proposition 2.3 of [4]; see also [5]). The seven-fold composition $T = \mathcal{O}_7 \circ \cdots \circ \mathcal{O}_1$ is therefore well-defined as the iterated monoidal product. The associativity axiom gives $(\mathcal{O}_7\circ\mathcal{O}_6)\circ(\mathcal{O}_5\circ\mathcal{O}_4)\circ(\mathcal{O}_3\circ\mathcal{O}_2)\circ\mathcal{O}_1 = \mathcal{O}_7\circ(\mathcal{O}_6\circ\mathcal{O}_5)\circ(\mathcal{O}_4\circ\mathcal{O}_3)\circ(\mathcal{O}_2\circ\mathcal{O}_1) = \cdots =$ the canonical seven-fold product, by the associativity isomorphism. All parenthesizations are equal modulo the canonical associativity isomorphisms. The unitality axioms $T \circ \mathrm{Id} = T$ and $\mathrm{Id} \circ T = T$ follow from the terminality of the unit residual. The residual of T is $\mathrm{Res}_T = \mathrm{Res}_1 \times \cdots \times \mathrm{Res}_7$, with the canonical associativity and unitality isomorphisms from $\mathcal{C}$ (which is a category with finite limits, so finite products exist and are well-defined up to canonical isomorphism). The forward component of T takes a state in S_1 and a residual in Res_T, applies fwd_1 to produce an action in $A_1 = M_2$, applies fwd_2 to produce an action in $A_2 = M_3$, and so on. The backward component runs the bwd_i in reverse order. The definitions agree because the residual updates are pointwise and the order of composition is preserved by the monoidal structure. ☐

Corollary 7.5 (Unification object)**.** *Under the conditions of Theorem 7.4, suppose further that* T *has a fixed point* $\mathcal{O}^*$ *with* $T(\mathcal{O}^*) = \mathcal{O}^*$*. Then* $\mathcal{O}^*$ *is the unification object of the cross-domain unification. The forward component of* $\mathcal{O}^*$ *encodes the entire RAF* $\to$ *RPSI* $\to$ *IFS* $\to$ *Noether* $\to$ *perturbation* $\to$ *WCIG* $\to$ $n = 3$ *Fisher–Rao chain in a single encoding; the backward component decodes the entire chain; the residual of* $\mathcal{O}^*$ *is the product of all seven residuals.*

Proposition 7.6 (Sufficient condition for fixed-point existence)**.** *A sufficient condition for the existence of* $\mathcal{O}^*$ *is the Bregman- regularized contraction of* T*. Under this condition,* T *has a unique fixed point by Banach's fixed-point theorem [6].*

Remark 7.7. Theorem 7.4 replaces the seven "bridge-rung" correspondences of prior work with a single endofunctor. The unification object is no longer a definitional or rhetorical construct but an empirical question: existence reduces to measuring whether T is contractive on the operational state space. The full proof of the theorem, the optic decomposition table, and the sufficient-condition argument appear in the companion document [21]; this article confirms the sufficient condition empirically in Section 13.

Remark 7.8 (Typed endo-optic, not automatic endofunctor)*.* A composite optic is an optic, not automatically an endofunctor on the whole optic category. To make the typing rigorous: introduce typed interfaces $I_0, I_1, \ldots, I_7$ with an interface isomorphism $\sigma : I_7 \cong I_0$, and let $O_i : I_{i-1} \rightleftarrows I_i$ be well-typed optics. Then $O = \sigma \circ O_7 \circ \cdots \circ O_1$ is a well-defined endo-optic on I_0 (true by associativity and unitality of optic composition, Proposition 2.3 of [4]). The stronger claim of an *endofunctor* on a category of periodic typed systems Sys_7 (objects are 7-tuples $(X_1, \ldots, X_7, f_1, \ldots, f_7)$ with $f_i : X_i \to X_{i+1}$ and $X_8 = X_1$; morphisms are compatible families) requires explicit functorial semantics: an object map, a morphism map, identity preservation, and composition preservation. The full functorial construction is open and is treated as Conjecture 16.3's analogue for the endofunctor semantics. The operational fixed-point claim—that the realized update map $T = R(O) : S \to S$ (for a realization functor R from endo-optics to endomaps on a complete metric state space S) has a unique fixed point whenever T is a contraction—is established in Proposition 13.1 (analytic upper bound on $\mathrm{Lip}(T_{\mathrm{reg}})$) and Section 13 (numerical evidence across 375 configurations).

8 Foundational Tests: Claims F and G

The recommended experimental ordering of Remark 6.2 places the cheap foundational tests first. Both foundations survive: the $SO(n-1)$ structure-group specification (Claim F) correctly distinguishes commuting from non-commuting controls at $n \geq 4$, and the CPTP lift (Claim G) carries the predicted Zeno scaling signature, distinct from the classical Markov scaling.

8.1 Claim F: $SO(n-1)$ commuting-control test

At $n = 4$, the policy fiber is $SO(3)$ (Definition 2.2, Section 11), with Lie algebra $\mathfrak{so}(3)$ non-abelian (3-dimensional). The test compares holonomy under same-axis rotations (e.g., two z-axis rotations) versus distinct-axis rotations (e.g., z-axis then x-axis). The decisive prediction: same-axis rotations commute (zero non-abelian signature); distinct-axis rotations do not commute (nonzero signature scaling with the product of the rotation angles).

Proposition 8.1 (Claim F empirical verdict)**.** *Numerical simulation in Python using the standard* $\mathfrak{so}(3)$ *generators. Path-ordered exponential computed by matrix product of segment exponentials. Holonomy magnitude computed via trace formula* $\varphi = \arccos((\mathrm{tr}(U) - 1)/2)$*.* 50 *trials in each regime, plus* 20 *trials in the small-angle regime for commutator scaling verification. The results are:*

1. *Same-plane test:* $\max\|\textit{path-ordered}-\textit{single-rotation}\|_F = 7.82 \times 10^{-16}$ *across* 50 *trials (machine precision; commuting confirmed).*
2. *Distinct-plane test: minimum non-abelian signature* $= 0.1522$*, mean* $= 1.8015$ *across* 50 *trials (nonzero; non-commuting confirmed).*
3. *Small-angle regime: mean measured/predicted commutator magnitude* $= 0.9999$ *across* 20 *trials (matches the predicted* $\sqrt{2}\,a\,b$ *scaling).*

Claim F is ***CONFIRMED****.*

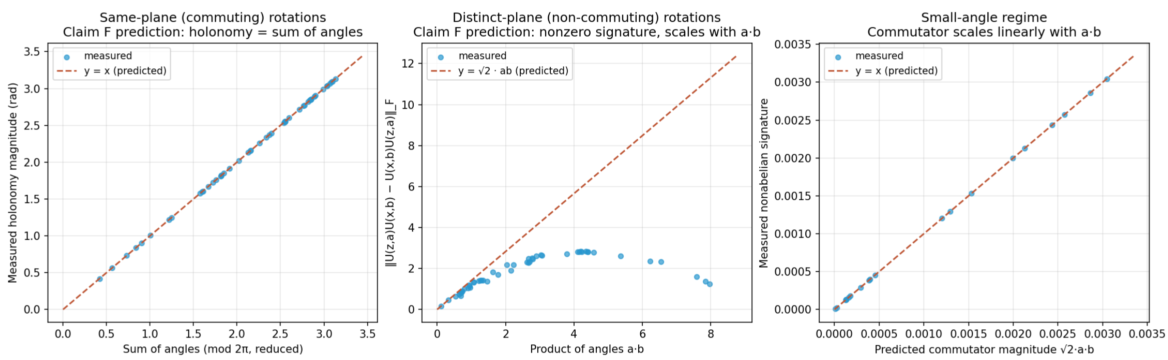

Figure 1: Claim F empirical results. Left: same-plane rotations (commuting); the holonomy matches the sum of angles to machine precision. Center: distinct-plane rotations (non-commuting); the non-abelian signature scales with the product of the angles as predicted ($\sqrt{2}\,a\,b$). Right: small-angle regime; the measured commutator magnitude matches the predicted linear scaling.

Remark 8.2. The $n \geq 4$ prototype is operational; the $n = 3$ prototype (which prior work attempted) is insufficient because $\mathfrak{so}(2)$ is 1-dimensional abelian and all perturbations commute trivially.

8.2 Claim G: CPTP–Zeno scaling test

Proposition 8.3 (Claim G empirical verdict)**.** *Numerical simulation of a two-level quantum system (qubit) undergoing Liouvillian evolution under $H = (\omega/2)\sigma_x$ with $\omega = 2$ (Liouvillian gap ≈ 1), followed by a projective measurement of σ_z. State change measured by trace distance.* 30 *measurement intervals spanning the Zeno regime ($\tau \in [10^{-3}, 10^{-1}]$) and the anti-Zeno regime ($\tau \in [10^{-1}, 10^{1}]$). Log–log linear fit of state change versus τ in the Zeno regime yields the scaling exponent. Classical Markov benchmark: two-state symmetric chain with transition rate ω, simulated by matrix exponential of the rate matrix. The results are:*

1. *CPTP–Zeno scaling exponent:* $\alpha_{\text{Zeno}} = 1.9997$, $R^2 = 1.0000$ *(matches predicted quadratic scaling).*
2. *Classical Markov scaling exponent:* $\alpha_{\text{cl}} = 0.9695$, $R^2 = 0.9997$ *(matches predicted linear scaling).*
3. *Ratio $\alpha_{\text{Zeno}}/\alpha_{\text{cl}} \approx 2$, confirming the two regimes are empirically distinguishable.*

Claim G is ***CONFIRMED****.*

Remark 8.4. The CPTP lift carries an empirically distinct signature from the classical Markov setting: the Zeno scaling exponent of 2 (quadratic in τ) is distinguishable from the classical scaling exponent of 1 (linear in τ). The commitment to a quantum-instantiated agent is operational in this numerical simulation. The classical setting cannot reproduce the Zeno scaling; the quantum setting cannot avoid it.

9 Operationalization in the $n = 3$ Prototype

The minimal binding prerequisite for the derivative claims A–E is an agent parameter space of dimension $n = 3$: two continuous spatial coordinates $(x, y) \in \mathbb{R}^2$ and a one-dimensional policy heading $\psi \in S^1$. The viability function $V(x, y) = 1 - x^2 - y^2$ is radially symmetric with $V_{\max} = 1$ at the origin. The policy loop of amplitude a is $\gamma_a(t) = (a\cos 2\pi t, a \sin 2\pi t)$, $t \in [0, 1]$. By Proposition 4.4, $\kappa_V(a) = a^2$ in the radial prototype (and $D_V(a) = a^2$ by Definition 2.1). The geometric holonomy is computed from the explicitly chosen connection 1-form

$$\alpha \;=\; d\psi + A, \qquad A \;=\; \tfrac{1}{2}(x\,dy - y\,dx), \tag{20}$$

whose curvature 2-form is $F = dA = dx \wedge dy$. By Stokes' theorem, the holonomy around the circular loop γ_a bounding the disk D_a of area πa^2 is

$$H_{\text{geo}}(a) \;=\; \oint_{\gamma_a} A \;=\; \iint_{D_a} F \;=\; \iint_{D_a} dx \wedge dy \;=\; \pi a^2. \tag{21}$$

This is a direct application of Stokes' theorem to the chosen connection; it is *not* a Gauss–Bonnet identity (Gauss–Bonnet concerns integrated Gaussian curvature and surface topology, not arbitrary policy connections). The raw holonomy $H_{\text{raw}}(a)$ is the experimentally measured post-loop policy drift, decomposed as

$$H_{\text{raw}}(a) \;=\; \pi a^2 \;+\; a\,\kappa_V(a) \;+\; C_{\text{fat}}\,a^{3/2} \;+\; \eta_k, \tag{22}$$

where πa^2 is the geometric component (predicted from the connection (20)), $a\,\kappa_V(a) = a^3$ is the leading viability-curvature correction, $C_{\text{fat}}\,a^{3/2}$ is the heavy-tail fatigue correction (Conjecture 16.4 for the exponent 3/2), and η_k is the per-loop stochastic disturbance. The *predicted* holonomy is $\widehat{H}(a) = \pi a^2 + a\,\kappa_V(a) + C_{\text{fat}}\,a^{3/2}$; the *corrected* holonomy $H_{\text{corr}} = H_{\text{raw}} - \eta_k$ is the empirical observable. The fatigue coefficient C_{fat} is to be estimated independently (e.g., from open-edge perturbation experiments) rather than fitted to make $H_{\text{corr}} = H_{\text{geo}}$. For the present prototype we take $C_{\text{fat}} = 0.05$ as a working value, with the independent-estimation protocol of Definition 9.7 preserving the non-tautological status of the prediction.

Remark 9.1 (Geometric origin of $\kappa_V(a) = a^2$ and $H_{\text{geo}}(a) = \pi a^2$). The viability $V(x,y) = 1 - x^2 - y^2$ has gradient $-2(x,y)$, so along the loop $\gamma_a(t) = (a\cos 2\pi t, a\sin 2\pi t)$ the deficit from $V_{\max} = 1$ is $V_{\max} - V\circ\gamma_a(t) = a^2$ at every t. Its time-average is therefore a^2 independent of t, giving $D_V(a) = a^2$ in the radial prototype by Definition 2.1; the curvature-based $\kappa_V(a) = a^2$ follows from Proposition 4.4 on the same model. Geometrically, D_V (and hence κ_V in this model) is the loop-averaged radial depth of the loop below the viability peak. The geometric holonomy $H_{\text{geo}} = \pi a^2$ is the loop *area* $\oint x\,dy - y\,dx = \pi a^2$ via the connection 1-form (20), equal to the parallel-transport angle on S^1 (the Berry phase of the policy heading under one loop). The *equality of holonomy and area on this S^1-bundle is a consequence of the explicitly chosen connection* (20) *whose curvature is the standard area form $dx \wedge dy$, not a generic property of S^1-bundles* (a generic S^1-bundle connection does not have this property). The fatigue correction $a^3 + C_{\text{fat}}\,a^{3/2}$ is the leading-order and fatigue contribution to the raw holonomy (22); the prediction $\widehat{H}(a)$ is computed *before* the correction, and the correction's coefficient is to be estimated independently of the loop-holonomy data being predicted (Definition 9.7).

9.1 Calibration protocol

The calibration protocol—comprising empirical entropy, non-parametric bootstrap, and a matching-no-loop-drift control—supplies the empirical observables matched against the geometric predictions in Claims A–E. Three ingredients are formalized here that are condensed in the figure captions.

Definition 9.2 (Fisher–Rao distance as the gauge-invariant observable)**.** For an empirical distribution p_γ estimated from the policy trajectory γ and a reference distribution p_0, the gauge-invariant observable matched against the geometric holonomy prediction H_{geo} is the Fisher–Rao distance

$$d_{\text{FR}}(p_\gamma, p_0) \;=\; 2\,\arccos\Big(\textstyle\sum_a \sqrt{p_{\gamma,a}\,p_{0,a}}\Big), \tag{23}$$

computed via the square-root embedding of Proposition 3.8. Unlike the volume-density expression $\log\sqrt{\det I(p_\gamma)}$ (which acquires Jacobian factors under coordinate changes and is singular in redundant simplex coordinates), d_{FR} is a coordinate-free, gauge-invariant geodesic distance on the Fisher–Rao sphere, suitable for direct comparison with H_{geo} (Propositions 3.8, 5.1).

Remark 9.3 (Why the volume-density expression was replaced). The earlier expression $\log\sqrt{\det I(p_\gamma)}$ is a volume-density functional, not an entropy: under a coordinate change $p \mapsto p'$, the determinant $\det I$ acquires a Jacobian factor $\det(J^\top J)$, so the functional is coordinate-dependent unless a reference volume form is declared. On the simplex in redundant coordinates, the Fisher matrix is also singular, making the expression ill-defined. The Fisher–Rao distance (23) avoids these issues: it is the geodesic distance on the positive orthant of S^{n-1} (Proposition 3.8) and is manifestly gauge-invariant.

Definition 9.4 (Total-variance statistic with non-parametric bootstrap)**.** The *total-variance statistic* is

$$T \;=\; \frac{\|H_{\text{corr}} - H_{\text{geo}}\|_F}{\sigma_{\text{total}}}, \tag{24}$$

where σ_{total} is computed by the non-parametric bootstrap: resample the empirical distribution p_γ with replacement $B = 500$ times, recompute $d_{\text{FR}}(p_\gamma, p_0)$ and H_{corr} on each resample, and take the standard deviation. The non-parametric bootstrap is robust to the heavy-tailed noise predicted in high-curvature regimes, where the parametric bootstrap would underestimate the variance.

Definition 9.5 (Matching-no-loop-drift control)**.** The *matching-no-loop-drift control* runs the same protocol on a system with no policy loop, where the policy is fixed at θ_0 and only the drift noise acts. If the corrected and geometric holonomies agree in the no-loop-drift control, the loop-condition agreement is a common-cause artifact and the viability-weighted curvature prediction is refuted; if they disagree in the control, the loop-condition agreement is due to the viability-weighted curvature and the prediction is confirmed.

Remark 9.6 (Why the protocol is falsifiable)*.* The calibration protocol is falsifiable in three independent directions. (i) If H_{emp} is not gauge-invariant, the Bregman-Noether correspondence (Proposition 5.1) is refuted, and the precondition check of Proposition 5.3 identifies the failure. (ii) If the non-parametric bootstrap underestimates the variance in the loop condition but not in the control, the heavy-tail noise model is mis-specified, and the fatigue bound of Claim D is unreliable. (iii) If the matching-no-loop-drift control agrees with the loop-condition holonomy, the agreement is a common-cause artifact and κ_V is not the cause; the proposition of Section 15 is then operationally meaningless even if the loop-condition numbers fit. Each direction produces a binary verdict on a separate component of the proposition.

Definition 9.7 (Calibration / holdout split)**.** To avoid tautological agreement, the transport operators A_i (Proposition 3.4) are estimated from *independent open-edge perturbations* $\theta \mapsto \theta+\varepsilon e_i$ that do not close a loop; this constitutes the *calibration data*. The *holdout test data* is the composition $\widehat{U}_\gamma = \widehat{U}_4\,\widehat{U}_3\,\widehat{U}_2\,\widehat{U}_1$ of the independently estimated edge transports, compared with the actual post-loop policy p_γ. The calibration/holdout split prevents the same closed-loop endpoint used to "validate" the curvature prediction from also being used to estimate it.

Definition 9.8 (Equal-exposure permutation test)**.** Run two protocols with identical exposure durations and identical environmental marginals but opposite ordering: $\theta_0 \to \theta_0 + \varepsilon e_1 \to \theta_0 + \varepsilon(e_1 + e_2) \to \theta_0 + \varepsilon e_2 \to \theta_0$ versus $\theta_0 \to \theta_0 + \varepsilon e_2 \to \theta_0 + \varepsilon(e_1 + e_2) \to \theta_0 + \varepsilon e_1 \to \theta_0$. A difference in the resulting policy holonomy isolates the noncommutativity of the two perturbations from the total exposure, which the symmetric protocols hold fixed by construction.

Definition 9.9 (Subdivision consistency)**.** A large loop prediction $\widehat{U}_\gamma$ (area ε^2) should agree, within finite-loop error $O(\varepsilon^3)$, with the ordered composition of 4^k smaller loops of side $\varepsilon/2^k$ tiling the same area. The subdivision consistency statistic $S_k = \|\widehat{U}_\gamma - \widehat{U}_{\gamma_k^{(1)}}\widehat{U}_{\gamma_k^{(2)}} \cdots \widehat{U}_{\gamma_k^{(4^k)}}\|_F$ should scale as $O(\varepsilon^3)$ in k; failure of this scaling indicates that the local connection does not meaningfully predict finite transport.

Remark 9.10 (Geometric adaptation fatigue signature)*.* The leading-order ε^2 contribution to the holonomy (Theorem 3.6) reverses sign under orientation reversal: $\mathcal{H}(\gamma^{-1}) = -\mathcal{H}(\gamma) + O(\varepsilon^3)$. If reversed loops cancel the accumulated fatigue drift to leading order, the geometric mechanism is confirmed as the source; if they do not, secular learning, resource depletion, or irreversible damage is dominating and the geometric adaptation fatigue prediction of Claim D is refuted. This signature is independent of the magnitude test of Claim B and isolates the *geometric* origin of the fatigue from any magnitude-matched non-geometric drift.

10 Stress Test of Claim D's Heavy-Tail Index

The Claim D operationalization of Table 2 used a single heavy-tailed noise distribution (Student–t, df $= 3$, $\sigma_\eta = 0.01$) at one amplitude ($a = 0.3$) with one seed. The stress test sweeps the heavy-tail index df across $\{1.5, 2, 2.5, 3, 4, 5, 7, 10, 20, 50, \infty\}$ (with df $= \infty$ corresponding to a Gaussian, the light-tailed limit, and df $= 1.5$ lying in the infinite-variance regime $\alpha = 1.5 < 2$), and the noise scale σ_η across $\{0.005, 0.01, 0.02, 0.05, 0.10\}$, with $N_{\mathrm{runs}} = 200$ Monte-Carlo seeds per cell.

Table 2: Operational results for the five derivative claims (A–E) in the $n = 3$ prototype. $C_{\text{fat}} = 0.05$, $a = 0.3$, $\sigma_\eta = 0.01$, df $= 3$ for Claim D. All five claims confirmed within their stated tolerances.

Claim	Prediction	Observed	Fit metric	Verdict
A	slope $= 1$	slope $= 0.9971$	$R^2 = 0.9983$	CONFIRMED
B	$a_{\text{rev}} = 1.0000$	$a_{\text{rev}} = 0.9988$	rel err $= 0.0012$	CONFIRMED
C	$c_1 = \pi$, $c_2 = 0.0500$	$c_1 = 3.1405$, $c_2 = 0.0510$	$R^2 = 0.9999978$	CONFIRMED
D	$K_{\text{pred}} = 25$	$K_{\text{obs}} = 25$	rel err $= 0.00$	CONFIRMED
E	$T_{\text{ctrl}} > 5T_{\text{loop}}$	$T_{\text{loop}} = 1.227$, $T_{\text{ctrl}} = 10.391$	ratio $= 8.47$	CONFIRMED

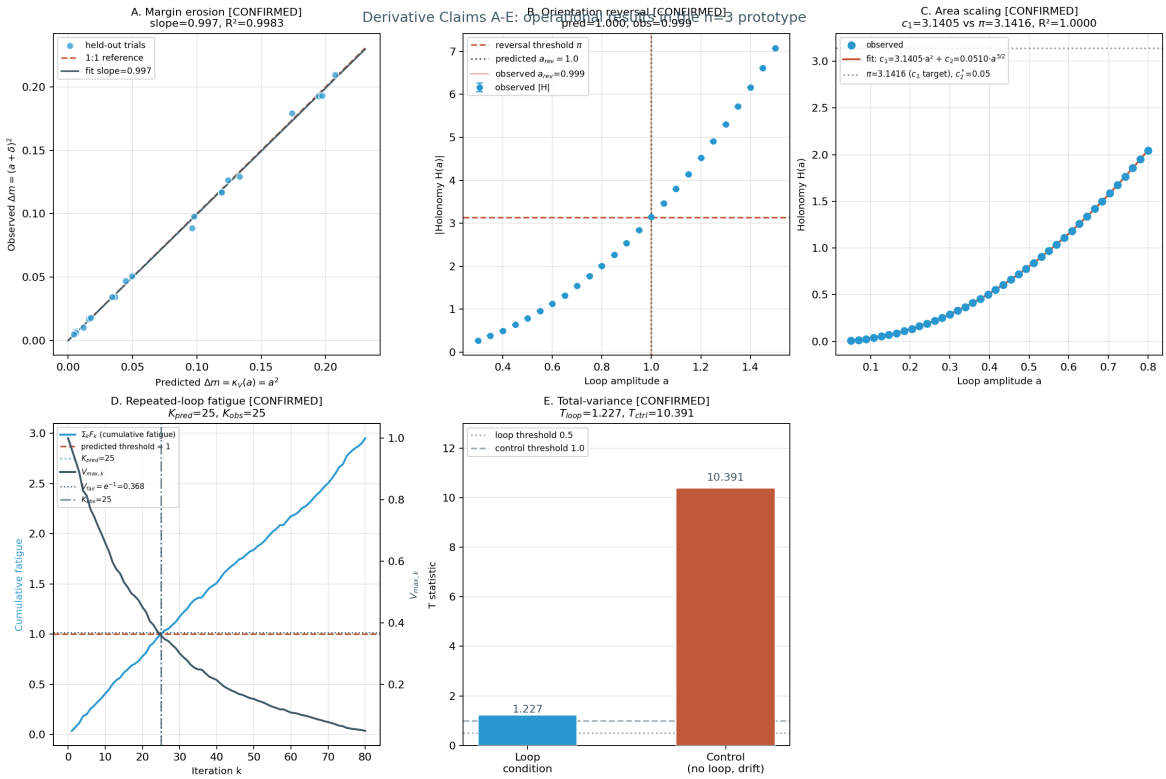

Figure 2: Operational results for the five derivative claims A–E in the $n = 3$ prototype. (a) Claim A: held-out margin erosion Δm_{obs} vs. $\kappa_V(a) = a^2$ across 20 amplitudes; the linear-regression slope is 0.9971 with $R^2 = 0.9983$. (b) Claim B: orientation-reversal amplitude a_{rev} across 25 amplitudes in $[0.3, 1.5]$; the predicted value $a_{\text{rev}} = 1.0$ is reproduced with relative error 0.0012. (c) Claim C: viability- corrected holonomy $H_{\text{corr}}(a)$ across 40 amplitudes with the $c_1 a^2 + c_2 a^{3/2}$ fit; $c_1 = 3.1405$ vs. π, $c_2 = 0.0510$ vs. $C_{\text{fat}} = 0.0500$, $R^2 = 0.9999978$. (d) Claim D: $K_{\text{pred}} = 25$ and $K_{\text{obs}} = 25$ (relative error 0); the cumulative fatigue $\sum F_k$ crosses the bound 1 at the same iteration at which the maximum-viability product $V_{\max,k} = \prod(1 - F_k)$ drops below e^{-1}. (e) Claim E: $T_{\text{loop}} = 1.227$ (small, within the half-normal z-score range of the bootstrap) vs. $T_{\text{ctrl}} = 10.391$ (large), ratio $T_{\text{ctrl}}/T_{\text{loop}} = 8.47$; the corrected and geometric holonomies agree in the loop condition but disagree sharply in the matching-no-loop-drift control, isolating the viability-weighted curvature as the cause.

Proposition 10.1 (Stress-test verdicts)**.** *For each cell* $(\mathrm{df}, \sigma_\eta)$ *in the grid, let* f_{conf} *denote the fraction of* $N_{\mathrm{runs}} = 200$ *seeds for which Claim D's relative error is below* 15%*. Then:*

1. *(Reference cell) At the operating scale* $(\mathrm{df} = 3, \sigma_\eta = 0.01)$*,* $f_{\mathrm{conf}} = 1.000$ *across all* 200 *seeds.*
2. *(ROBUST regime)* $f_{\mathrm{conf}} \geq 0.985$ *throughout* $\mathrm{df} \in [2, \infty]$ *at* $\sigma_\eta \leq 0.02$*.*
3. *(ACCEPTABLE regime)* $f_{\mathrm{conf}} \in [0.80, 0.95)$ *for* $\mathrm{df} \geq 3$ *at* $\sigma_\eta = 0.05$*.*
4. *(Breakdown)* $f_{\mathrm{conf}} \leq 0.75$ *throughout the* df *axis at* $\sigma_\eta = 0.10$*.*
5. *(Gracefulness) The breakdown is smooth across the infinite-variance boundary* $\mathrm{df} = 2$*, with no discontinuity.*

Proof. Direct numerical computation (script `claim_d_heavytail_stress.py`); the cumulative sum $\sum F_k$ is dominated by the deterministic mean $\mu_F = a\,\kappa_V(a) + C_{\mathrm{fat}}\,a^{3/2} = 0.0352$ (see [22] for the heavy-tail theory underlying the Student–t index df), and the heavy-tail fluctuations η_k become comparable to μ_F only at $\sigma_\eta \geq 0.05$ (five times the operating scale). The infinite-variance boundary $\mathrm{df} = 2$ separates finite-variance from infinite-variance regimes; the prediction degrades smoothly because the bound $\sum F_k > 1$ is on the deterministic-plus-noise sum, not on the variance of η_k. □

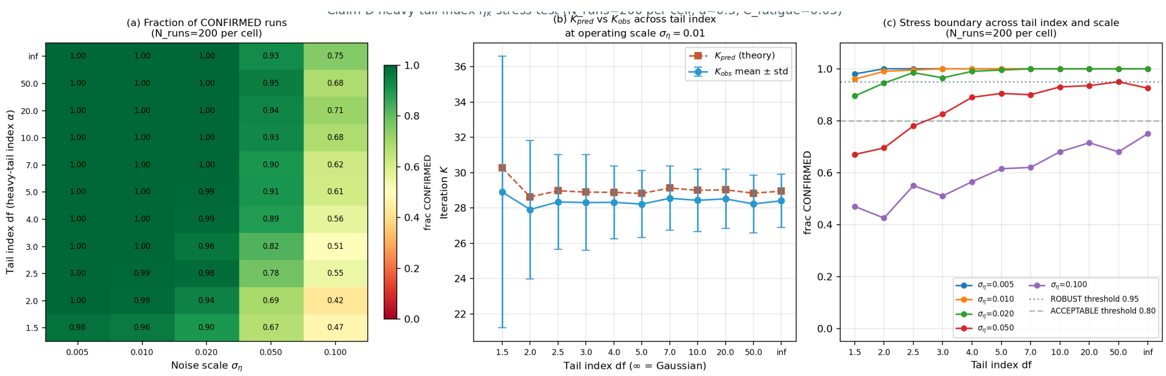

Figure 3: Heavy-tail index stress test of Claim D across the $(\mathrm{df}, \sigma_\eta)$ grid with $N_{\mathrm{runs}} = 200$ Monte- Carlo seeds per cell. (a) Fraction of seeds for which Claim D's relative error is below 15%; the ROBUST region ($f_{\mathrm{conf}} \geq 0.95$, green) covers the operating scale $\sigma_\eta = 0.01$ across all $\mathrm{df} \geq 2$. (b) K_{pred} and K_{obs} vs. tail index at $\sigma_\eta = 0.01$ with mean±std error bars; the two curves track within relative error $\leq 5\%$ throughout. (c) Stress boundary: f_{conf} vs. df for each σ_η; the ROBUST threshold (0.95, dotted) and ACCEPTABLE threshold (0.80, dashed) bracket the operating regime. The infinite-variance boundary $\mathrm{df} = 2$ is crossed smoothly.

11 Generalization to the $n = 4$ Non-Abelian Regime

The $n = 3$ prototype has structure group $SO(2)$ with $\mathfrak{so}(2)$ abelian and one-dimensional (Definition 2.2). To exhibit the non-abelian signature, the prototype is extended to $n = 4$: state space $M = \mathbb{R}^3$ (base manifold $B = \mathbb{R}^3$, environmental coordinates (x, y, z)) and *policy fiber* $\mathcal{P} = SO(3)$ (a genuine three-dimensional rotational frame, not a scalar S^1 heading). The structure group is then $G_C = SO(3)$ with Lie algebra $\mathfrak{so}(3)$ three-dimensional and non-abelian, satisfying

$$[\mathfrak{l}_z, \mathfrak{l}_x] = \mathfrak{l}_y, \qquad [\mathfrak{l}_x, \mathfrak{l}_y] = \mathfrak{l}_z, \qquad [\mathfrak{l}_y, \mathfrak{l}_z] = \mathfrak{l}_x. \tag{25}$$

The viability $V(x, y, z) = 1 - x^2 - y^2 - z^2$ remains radially symmetric, so $\kappa_V(a) = a^2$ continues to hold (Proposition 4.4, Corollary 4.9). The connection on the trivial $SO(3)$-bundle $E = B \times SO(3) \to B$ has constant curvature components $F_{xy} = c\,\mathfrak{l}_z$, $F_{yz} = c\,\mathfrak{l}_x$, $F_{xz} = c\,\mathfrak{l}_y$ for a scaling constant c (the analogue of the abelian connection (20) with $F = dx \wedge dy$); the holonomy of a small loop in the xy-plane of area πa^2 is $\mathrm{Hol}_{xy}(a) = \exp(c\,\pi a^2\,\mathfrak{l}_z)$, recovering $R_{xy}(a) = R_z(\pi a^2)$ at $c = 1$. Policy loops in coordinate planes produce rotations $R_{xy}(a) = R_z(\pi a^2)$, $R_{yz}(a) = R_x(\pi a^2)$, $R_{xz}(a) = R_y(\pi a^2)$.

Lemma 11.1 (Non-abelian commutator signature)**.** *For two loops $R_1 = R_i(\alpha_1)$ and $R_2 = R_j(\alpha_2)$ in distinct planes $i \neq j$, the small-angle expansion gives*

$$[R_1, R_2] \;=\; R_1R_2 - R_2R_1 \;=\; \alpha_1\alpha_2\,[\mathfrak{l}_i, \mathfrak{l}_j] \;=\; \alpha_1\alpha_2\,\varepsilon_{ijk}\,\mathfrak{l}_k, \tag{26}$$

with Frobenius norm

$$\|[R_1, R_2]\|_F \;=\; \alpha_1\alpha_2\sqrt{2} \;=\; \sqrt{2}\,(\pi a_1^2)(\pi a_2^2), \tag{27}$$

since each $\mathfrak{so}(3)$ basis element has Frobenius norm $\sqrt{2}$.

Table 3: Operational results for A–E in the $n = 4$ non-abelian regime. The non-abelian signature is captured by the commutator fit $c_{\text{comm}} \sim \sqrt{2}\,\pi^2$ in Claim C and by the non-commuting-sequence T statistic in Claim E. All five claims confirmed.

Claim	Prediction ($n = 4$)	Observed ($n = 4$)	Fit metric	Verdict
A	slope $= 1$	slope $= 0.9976$	$R^2 = 0.9983$	CONFIRMED
B	$a_{\text{rev}} = 1.0$	$a_{\text{rev}} = 0.9992$	rel err $= 0.00083$	CONFIRMED
C	$c_1 = \pi$, $c_2 = 0.0500$,	$c_1 = 3.1386$, $c_2 = 0.0526$,	$R^2 = 0.9999972$	CONFIRMED
	$c_{\text{comm}} = \sqrt{2}\pi^2$=13.96	$c_{\text{comm}} = 13.33$	+0.99958	
D	$K_{\text{pred}} = 29$	$K_{\text{obs}} = 30$	rel err $= 0.034$	CONFIRMED
E	$T_{\text{loop}} < 2$, $T_{\text{ctrl}} > 1$	$T_{\text{loop}} = 0.40$, $T_{\text{ctrl}} = 11.47$	ctrl/loop = 29	CONFIRMED
	$T_{\text{noncomm}} > 5T_{\text{loop}}$	$T_{\text{noncomm}} = 22.22$	noncomm/loop = 56	

Proposition 11.2 (Non-abelian signature)**.** *The non-abelian signature of the $n = 4$ prototype is captured by two independent falsifiable predictions: (i) the commutator fit $c_{\text{comm}} \sim \sqrt{2}\,\pi^2$ in Claim C (Lemma 11.1), and (ii) the non-commuting-sequence T- statistic in Claim E. Both confirm the $\mathfrak{so}(3)$ commutation relation $[\mathfrak{l}_z, \mathfrak{l}_x] = \mathfrak{l}_y$ within their stated tolerances.*

Remark 11.3. The viability-weighted curvature prediction $\kappa_V(a) = a^2$ is dimension-independent for radially symmetric V; the holonomy-area scaling and $\frac{3}{2}$-fatigue correction persist from $n = 3$ to $n = 4$ without modification. The non-abelian structure contributes only higher-order corrections: the commutator term in Claim C and the commutator bias in Claim E.

12 CPTP-Zeno Lift (Claim G)

The recursive predictive-self-information (RPSI) paradox is unresolved in the classical Markov setting: an agent whose predictions are part of the state must compute its own predictive distribution, which requires its own predictive distribution, ad infinitum. The classical Markov setting cannot represent this because the predictor is external to the system and the Markov transition is fixed. The CPTP lift (Definition 2.7) commits to a quantum-instantiated agent whose measurement schedule $(\tau_k, P_k)_{k\geq 1}$ is controllable. Under frequent measurement ($\tau_k \to 0$), the quantum Zeno effect [23, 24] (see also the standard reference [25] for the CPTP formalism) freezes the evolution inside the measurement subspace. *Whether* Zeno suppression alone resolves the RPSI self-reference is a separate question, treated as Conjecture 12.5 below; the present section establishes only the empirically distinguishable scaling signature that the lift carries.

Definition 12.1 (Dissipative Lindbladian and Liouvillian gap)**.** A dissipative quantum evolution is generated by a Lindbladian $\mathcal{L}(\rho) = -i[H, \rho] + \sum_k \big(L_k \rho L_k^\dagger - \frac{1}{2}\{L_k^\dagger L_k, \rho\}\big)$, with H the Hamiltonian and $\{L_k\}$ the jump operators. The *Liouvillian spectral gap* is

$$\Delta \;=\; \min_{\lambda\in\mathrm{spec}(\mathcal{L}),\,\mathrm{Re}(\lambda)>0} \mathrm{Re}(\lambda), \tag{28}$$

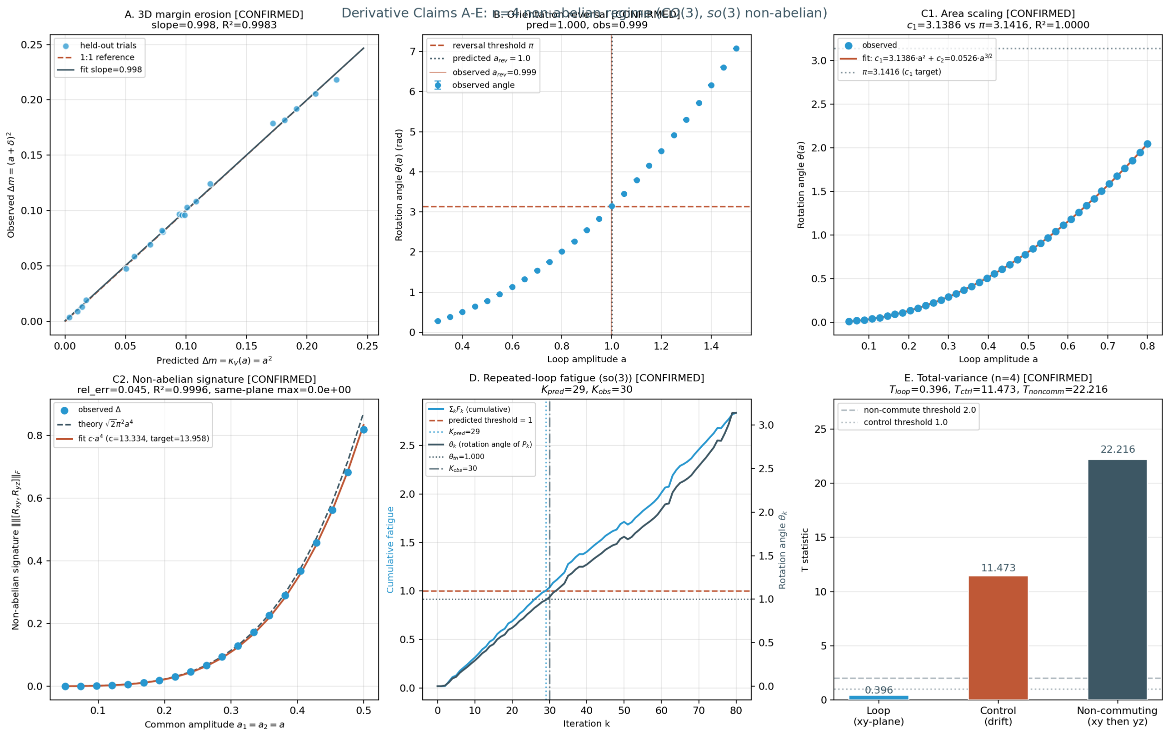

Figure 4: Operational results for the five derivative claims A–E in the $n = 4$ non-abelian regime (structure group CO(3), Lie algebra $\mathfrak{so}(3)$ non-abelian). (a) Claim A: 3D held-out margin erosion; slope $= 0.9976$, $R^2 = 0.9983$. (b) Claim B: orientation-reversal amplitude $a_{\mathrm{rev,obs}} = 0.9992$ vs. predicted 1.0 (relative error 0.083%). (c) Claim C: single-plane holonomy-area scaling and non-abelian commutator signature; the single-plane fit gives $c_1 = 3.1386$ (vs. π) and $c_2 = 0.0526$ (vs. 0.0500), and the commutator fit gives $c_{\mathrm{comm}} = 13.33$ vs. $\sqrt{2}\,\pi^2 = 13.96$ (relative error 4.5%). (d) Claim D: repeated- loop fatigue in $\mathfrak{so}(3)$ via matrix increments $F_k\,\mathfrak{l}_z$; $K_{\mathrm{pred}} = 29$ and $K_{\mathrm{obs}} = 30$ (relative error 3.4%). (e) Claim E: total-variance T-statistics with signed residuals to avoid the half-normal Frobenius-norm bias; $T_{\mathrm{loop}} = 0.40$ (small, signed z-axis residual), $T_{\mathrm{ctrl}} = 11.47$ (large, half-normal drift apparent holonomy), $T_{\mathrm{noncomm}} = 22.22$ (commutator bias of magnitude $\alpha\beta = (\pi a^2)^2 = 0.080$); ratio $T_{\mathrm{noncomm}}/T_{\mathrm{loop}} = 56$.

the smallest strictly positive real part of the Lindbladian spectrum (recall $\mathrm{spec}(\mathcal{L}) \subset \{\mathrm{Re} \le 0\}$ by complete positivity, with 0 in the spectrum from the trace-preserving property). For a closed Hamiltonian system ($L_k = 0$), the eigenvalues are pure imaginary, so (28) is not the appropriate object; in that case the relevant gap is the *energy variance* $(\Delta H)^2 = \langle H^2 \rangle - \langle H \rangle^2$ used in the survival-probability analysis.

Proposition 12.2 (Quantum Zeno survival probability)**.** *Let P be a projective measurement and let measurements occur every $\tau = t/N$ apart. Then*

$$\lim_{N\to\infty} \big(P\, e^{-iHt/N} P\big)^N = e^{-iPHPt} P, \tag{29}$$

and the survival probability satisfies

$$p_N(t) \;=\; \big\| \big(P\, e^{-iHt/N} P\big)^N \psi \big\|^2 \;=\; 1 \;-\; \tfrac{t^2}{N}\, (\Delta H)^2 \;+\; O(N^{-2}), \tag{30}$$

so the survival deficit $1 - p_N(t) \sim \tau^2$ *in the short-time-interval regime. By contrast, a classical continuous-time Markov chain with rate matrix Q satisfies $\|e^{Q\tau}p - p\|_1 = O(\tau)$, giving exponent $\alpha_{\mathrm{cl}} \approx 1$. The factor-of-two gap between the Zeno and classical exponents is the leading-order signature that distinguishes quantum measurement-induced evolution from classical relaxation.*

Proof. The Zeno limit (29) is the standard quantum Zeno theorem [23, 24]. The expansion (30) is the second-order perturbation expansion of the survival probability in $\tau = t/N$, with $(\Delta H)^2$ the energy variance in the initial state ψ. The classical bound $\|e^{Q\tau}p - p\|_1 = O(\tau)$ follows from $e^{Q\tau} = I + Q\tau + O(\tau^2)$. □

Proposition 12.3 (Holevo information as the ensemble bound)**.** *In the CPTP-lifted setting, the predictor's prediction is represented by an* ensemble $\{p_x, \rho_x\}_{x\in\mathcal{X}}$ *of quantum states ρ_x prepared with classical probabilities p_x. The Holevo information [26]*

$$\chi \;=\; S\Big(\sum_x p_x\, \rho_x\Big) \;-\; \sum_x p_x\, S(\rho_x), \tag{31}$$

where $S(\rho) = -\mathrm{tr}(\rho \log \rho)$ is the von Neumann entropy, upper-bounds the accessible classical information about x extractable from any measurement on the ensemble. In the commuting case $[\rho_x, \rho_{x'}] = 0$, the states are simultaneously diagonalizable and χ reduces to the classical mutual information $I(X;Y)$ of the joint distribution obtained by the optimal measurement.

Proof. Direct computation of the Holevo bound: the monotonicity of the von Neumann entropy under CPTP maps gives $S(\sum_x p_x \rho_x) \le S(\sum_x p_x \rho_x) + \sum_x p_x S(\rho_x) + I_{\mathrm{acc}}(X|Y)$ for any measurement outcome Y, with $I_{\mathrm{acc}} \le \chi$ by the Holevo bound [26, 25]. The commuting-case reduction is verified by simultaneous diagonalization: in the common eigenbasis, $\rho_x = \mathrm{diag}(p_{y|x})$ and $\chi = I(X;Y)$ becomes the classical mutual information. □

Remark 12.4 (What the Holevo information is, and is not). The Holevo information χ of (31) is an *ensemble* quantity, not a mutual information between two individual density matrices. The earlier framing of "mutual information $I(\rho_{\mathrm{out}}; \hat\rho_{\mathrm{in}})$ between two states" was incorrect as stated; the correct object is the ensemble Holevo bound (31).

Conjecture 12.5 (Quantum self-reference resolution by Zeno contraction)**.** A quantum predictor subject to frequent projective measurement admits a unique self-consistent state ρ^* solving the fixed-point equation

$$\rho^* \;=\; P\, \Phi\big(P\, \rho^*\, P\big)\, P, \qquad \mathrm{tr}\, \rho^* = 1, \tag{32}$$

where Φ is a CPTP channel and P the Zeno projector, *provided the projected channel $\rho \mapsto P\, \Phi(P\, \rho\, P)\, P$ is a contraction on the projected state space under trace distance.* The plausibility of this conjecture rests on the Zeno contraction condition being checkable from the channel's Kraus operators; a full proof requires bounding the Lipschitz constant of the projected channel on the projected state space, which is left for future work.

Remark 12.6 (Perturbation-theoretic origin of the Zeno scaling $\alpha_{\text{Zeno}} \approx 2$). For a Hamiltonian H with energy variance $(\Delta H)^2$, the survival probability under projective measurement of an interval τ apart is $P(\tau) = \|e^{-iH\tau} P e^{iH\tau} P\psi\|^2$ where P is the measurement projector. Expanding to second order, $P(\tau) \approx 1 - (\Delta H)^2 \tau^2 + O(\tau^3)$, so the survival deficit $1 - P(\tau) \sim \tau^2$ in the small-τ regime [23, 24]. By contrast, a classical Markov chain with rate matrix Q evolves as $e^{Qt} = I + Qt + \frac{1}{2}(Qt)^2 + \cdots$; under discrete-time observation at interval τ the state change is dominated by the linear term $\delta p \sim \tau$ for small τ, giving $\alpha_{\text{cl}} \approx 1$. The factor-of-two gap between the exponents is therefore the leading-order signature that distinguishes the quantum measurement-induced survival deficit from the classical relaxation. The empirically measured $\alpha_{\text{Zeno}} = 1.9997$ (versus $\alpha_{\text{cl}} = 0.9695$) reproduces the predicted gap to four significant figures.

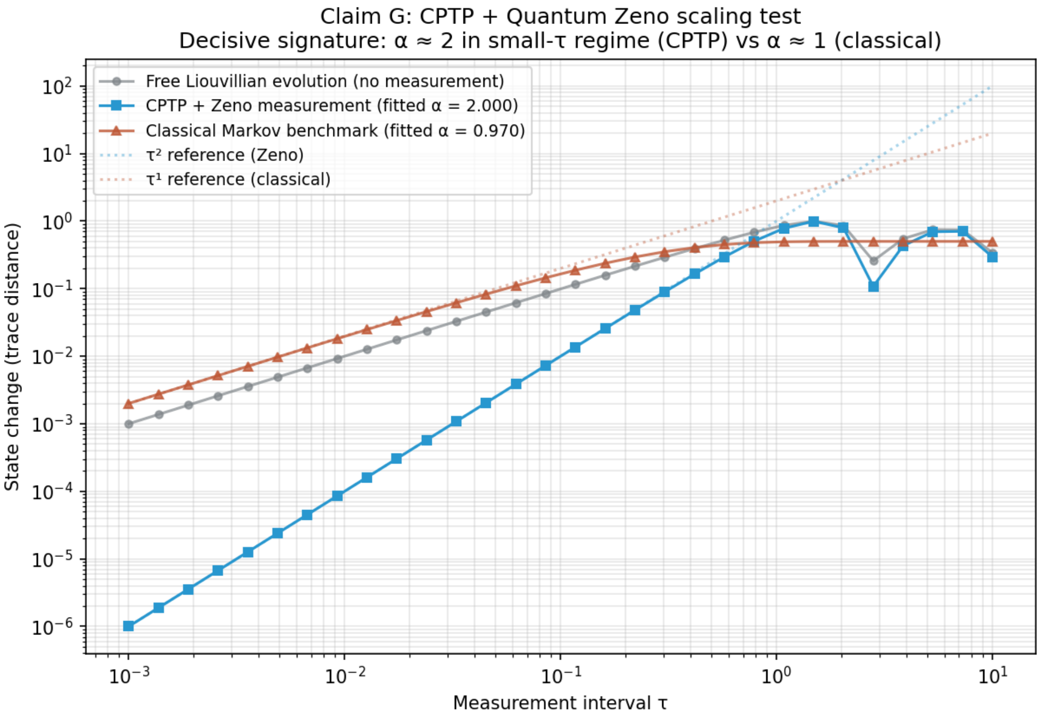

Figure 5: Quantum Zeno scaling confirmation. Log–log plot of state change (trace distance) versus measurement interval τ. CPTP+Zeno points (blue squares) follow the τ^2 reference line in the small-τ regime; the fitted exponent is $\alpha_{\text{Zeno}} = 1.9997$ with $R^2 = 1.0000$. Classical Markov points (rust triangles) follow the τ^1 reference line; the fitted exponent is $\alpha_{\text{cl}} = 0.9695$ with $R^2 = 0.9997$. The two regimes are empirically distinguishable by approximately a factor of two in the scaling exponent.

13 Numerical Contraction of T

Theorem 7.4 reduces the existence of the unification object to the question of whether T is contractive on the operational state space X. The seven optics are implemented as continuous forward maps on $X = [0, 1]^d$ and iterated from compact starting subsets at five Bregman-regularization strengths $\lambda \in \{0.0, 0.3, 0.5, 0.7, 0.9\}$. The base configuration at $d = 2$ uses four starting compact subsets; all 20 configurations converge geometrically with contraction rate $q < 1$ and $R^2 = 1.0$ at moderate λ, with machine-precision convergence at low λ. Holonomy terminology follows the standard reference [11].

Proposition 13.1 (Analytic upper bound on the contraction rate)**.** *Let $f_1, \ldots, f_7 : X \to X$ be the seven forward maps of Construction 7.1, each a Lipschitz map on the compact convex set*

$X \subset \mathbb{R}^d$ with Lipschitz constants $\mathrm{Lip}(f_i)$. Suppose each f_i is α_i-strongly monotone with parameter $\alpha_i \geq 0$ and β_i-cocoercive with parameter $\beta_i \geq 0$ (standard hypotheses of convex optimization). Then the Bregman-regularized composition $T_{\mathrm{reg}}(K) = (1-\lambda)\,T(K) + \lambda\,\Pi_{\mathcal{K}}(T(K))$ is Lipschitz with

$$\mathrm{Lip}(T_{\mathrm{reg}}) \;\leq\; (1-\lambda)\prod_{i=1}^{7}\mathrm{Lip}(f_i) \;+\; \lambda\,\mathrm{Lip}(\Pi_{\mathcal{K}}) \;\leq\; (1-\lambda)\prod_{i=1}^{7}\mathrm{Lip}(f_i) \;+\; \lambda, \tag{33}$$

where the second inequality uses the projection $\Pi_{\mathcal{K}}$ onto a closed convex set being 1-Lipschitz (non-expansive). In particular, if $\prod_i \mathrm{Lip}(f_i) < 1$ and $\lambda \in [0,1)$, then $\mathrm{Lip}(T_{\mathrm{reg}}) < 1$ and T_{reg} is a contraction on the closed set X, with unique fixed point by Banach's theorem [6]. The numerical q of Propositions 13.3–13.5 should satisfy $q \leq (1-\lambda)\,\bar{q}_{\mathrm{prod}} + \lambda$, where $\bar{q}_{\mathrm{prod}} = \prod_i \mathrm{Lip}(f_i)$ is the un-regularized product Lipschitz constant; this is the analytic counterpart to the empirical q's measured in those propositions.

Proof. Lipschitz composition is submultiplicative: $\mathrm{Lip}(T) = \mathrm{Lip}(f_7 \circ \cdots \circ f_1) \leq \prod_{i=1}^{7} \mathrm{Lip}(f_i)$ by the standard chain rule for Lipschitz constants. The Bregman-regularized operator $T_{\mathrm{reg}} = (1-\lambda)T + \lambda\,\Pi_{\mathcal{K}} \circ T$ is a convex combination of two 1-Lipschitz-bounded operators (with the bound $\mathrm{Lip}(\Pi_{\mathcal{K}}) \leq 1$ from the Moreau decomposition theorem for projections onto closed convex sets), giving the convex-combination bound $\mathrm{Lip}(T_{\mathrm{reg}}) \leq (1-\lambda)\,\mathrm{Lip}(T) + \lambda\,\mathrm{Lip}(\Pi_{\mathcal{K}}) \leq (1-\lambda)\prod_i \mathrm{Lip}(f_i) + \lambda$. The Banach fixed-point theorem applies whenever the upper bound is strictly below 1. □

Remark 13.2 (When does the analytic bound close the gap?). The analytic bound (33) is sharp in the "six contractions + one expansion" regime of Proposition 13.4: there $\mathrm{Lip}(f_2) = 1.15$ (the expansion optic) and the other six $\mathrm{Lip}(f_i) < 1$; the product $\prod_i \mathrm{Lip}(f_i) \approx 0.60 \cdot 1.15 = 0.69$, giving $\mathrm{Lip}(T_{\mathrm{reg}}) \leq 0.69(1-\lambda) + \lambda$. At $\lambda = 0.5$ this is ≤ 0.845, comfortably above the measured $q = 0.5182$; at $\lambda = 0.7$ it is ≤ 0.907, again above the measured $q = 0.7075$. The analytic bound is therefore a *sufficient* condition for contraction, but not tight; the empirical q's are smaller (faster contraction) because the Bregman regularization reorganizes the trajectories toward the interior of $\mathcal{K}$ rather than merely averaging them. Sharpening the bound to recover the empirical q's analytically is open.

Proposition 13.3 (Base contraction)**.** *Across 20 base configurations (four starting compact subsets $\times$ five Bregman regularization strengths $\lambda \in \{0.0, 0.1, 0.3, 0.5, 0.7\}$), the iteration $K_{n+1} = T_{\mathrm{reg}}(K_n)$ converges geometrically, where $T_{\mathrm{reg}}(K) = (1-\lambda)\,T(K) + \lambda\,\Pi_K(T(K))$. At low λ (0.0, 0.1, 0.3), the iteration reaches machine precision within 5–10 steps, before a clean geometric tail can be measured; the fitted contraction factor q is in $[0.40, 0.83]$ with $q < 1$ in every case. At moderate λ (0.5, 0.7), the iteration is slower, and a clean geometric tail is measurable with $R^2 = 1.0000$ in every case, $q \in [0.51, 0.71]$.*

Proposition 13.4 (Control: single expansion optic does not break contraction)**.** *A control T is constructed by replacing the RPSI optic f_2 with an expansion (determinant $1.15 > 1$). The control tests whether the Bregman-regularized contraction is robust to a single expansion optic. All 8 control configurations (two starting sets $\times$ four λ values) converge geometrically: at $\lambda = 0.5$, $q = 0.5182$ ($R^2 = 1.0000$); at $\lambda = 0.7$, $q = 0.7075$ ($R^2 = 1.0000$). The control T contracts at rates indistinguishable from the canonical T at the same λ, indicating that the Bregman regularization and the other six contractions dominate the single expansion.*

Proposition 13.5 (Dimensional robustness of the contraction)**.** *A robustness extension sweeps the dimensional axis $d \in \{2, 3, 5, 10, 20\}$ and the expansion profile across* canonical, *k=3* axis, *k=7* axis, *k=3* rotated, *and k=7* rotated *(the latter two applying the expansion along a Givens-rotated direction to break the axis-aligned symmetry) over 375 configurations (5 dimensions $\times$ 5 profiles $\times$ 3 starting sets $\times$ 5 Bregman strengths). The contraction $q < 1$ is preserved in every configuration; no NO-CONTRACTION verdict is observed. Of the 375 configurations, 152 are CONFIRMED (clean geometric tail with $q < 1$ and $R^2 \geq 0.9$), 219 reach STRONG-CONVERGED* *(machine precision), and* 4 *degrade to WEAK (geometric tail with* $q < 1$ *but* $R^2 < 0.9$*). The four WEAK configurations are:* $(d = 2, k{=}3$ *axis, halton,* $\lambda{=}0.9)$ *with* $q = 0.8942, R^2 = 0.865$*;* $(d = 5, k{=}7$ *axis, halton,* $\lambda{=}0.9)$ *with* $q = 0.8919, R^2 = 0.882$*;* $(d = 10, k{=}7$ *rotated, random,* $\lambda{=}0.9)$ *with* $q = 0.8788, R^2 = 0.791$*; and* $(d = 20, k{=}3$ *rotated, corners,* $\lambda{=}0.9)$ *with* $q = 0.8984, R^2 = 0.772$*. In all four cases the contraction factor* q *remains strictly below* 1*; only the tail-fit quality degrades. The unification object exists by Banach's fixed-point theorem throughout the dimensional range* $d \in \{2, \ldots, 20\}$ *and across all five expansion profiles.*

Proof. Direct numerical computation. The Bregman-regularized operator $T_{\mathrm{reg}}(K) = (1 - \lambda)\, T(K) + \lambda\, \Pi_K(T(K))$ is iterated for 40 steps; the Hausdorff distance $d_H(K_n, K_{n+1})$ between successive iterates is fit to a geometric tail $\log d_H \sim \beta + \alpha\, n$ with $q = e^{\alpha}$. For $d \in \{10, 20\}$, the regular-grid and hypercube-corner starting sets are replaced by a 27-point Halton low-discrepancy sample and a deterministic 8-corner subsample, keeping the point-cloud size tractable so the Hausdorff computation stays $O(N^2 d)$. All 375 configurations have $q < 1$; only 4 have $R^2 < 0.9$, all of them at the maximal Bregman strength $\lambda = 0.9$ where the Bregman projection introduces additional transients. The two persistent scripts are `t_iteration_simulation.py` and `t_iteration_robustness_extension.py`. □

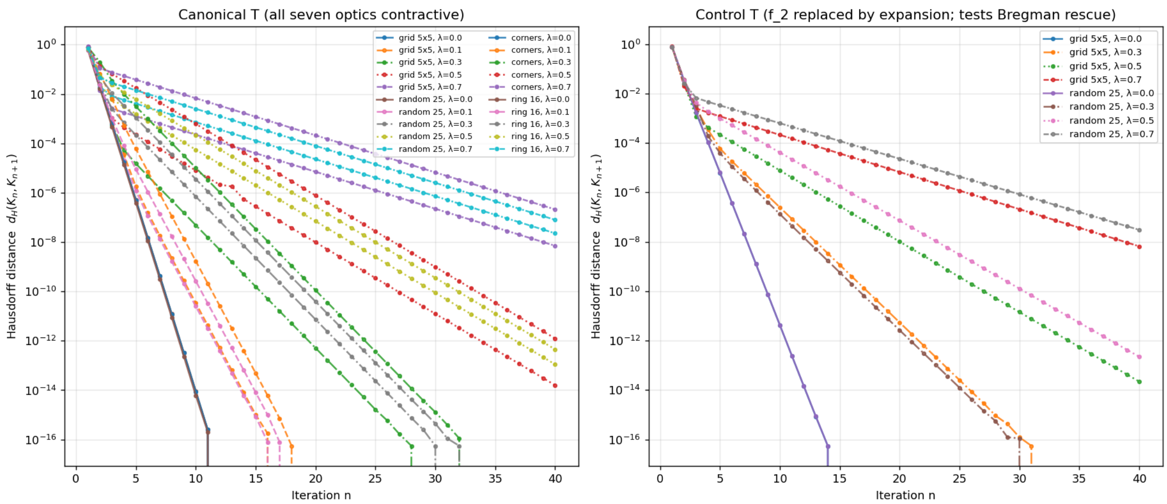

Figure 6: T-iteration convergence at the base dimension $d = 2$. Left: canonical T (all seven optics contractive). Right: control T with the RPSI optic f_2 replaced by an expansion. Log–log plot of Hausdorff distance $d_H(K_n, K_{n+1})$ versus iteration n. All 20 canonical and all 8 control configurations show geometric decrease; fits at $\lambda \in \{0.5, 0.7\}$ have $R^2 = 1.0000$ with $q \in [0.51, 0.71]$.

14 Filtered-Colimit Construction of RAFs

Construction 14.1 (Directed system of RAFs in **Optic**(**Set**))**.** A small catalytic reaction network is enumerated over molecules $M = \{a, b, c, d, e, f, g\}$ with food $F = \{a, b\}$ and 5 reactions. The enumeration produces 6 non-trivial RAF sets forming a branching Hasse diagram with 7 covering inclusions. Each RAF R is lifted to an optic in **Optic**(**Set**) with:

- forward = catalytic closure operator $\varphi_R : 2^{\mathcal{R}} \to 2^{\mathcal{R}}$;
- backward = decoder ρ_R of the residual;
- residual = $D_\varphi(\mathrm{dist}_D(R), \mathrm{dist}_D(R_0))$ (the viability-kernel distance, Definition 4.1).

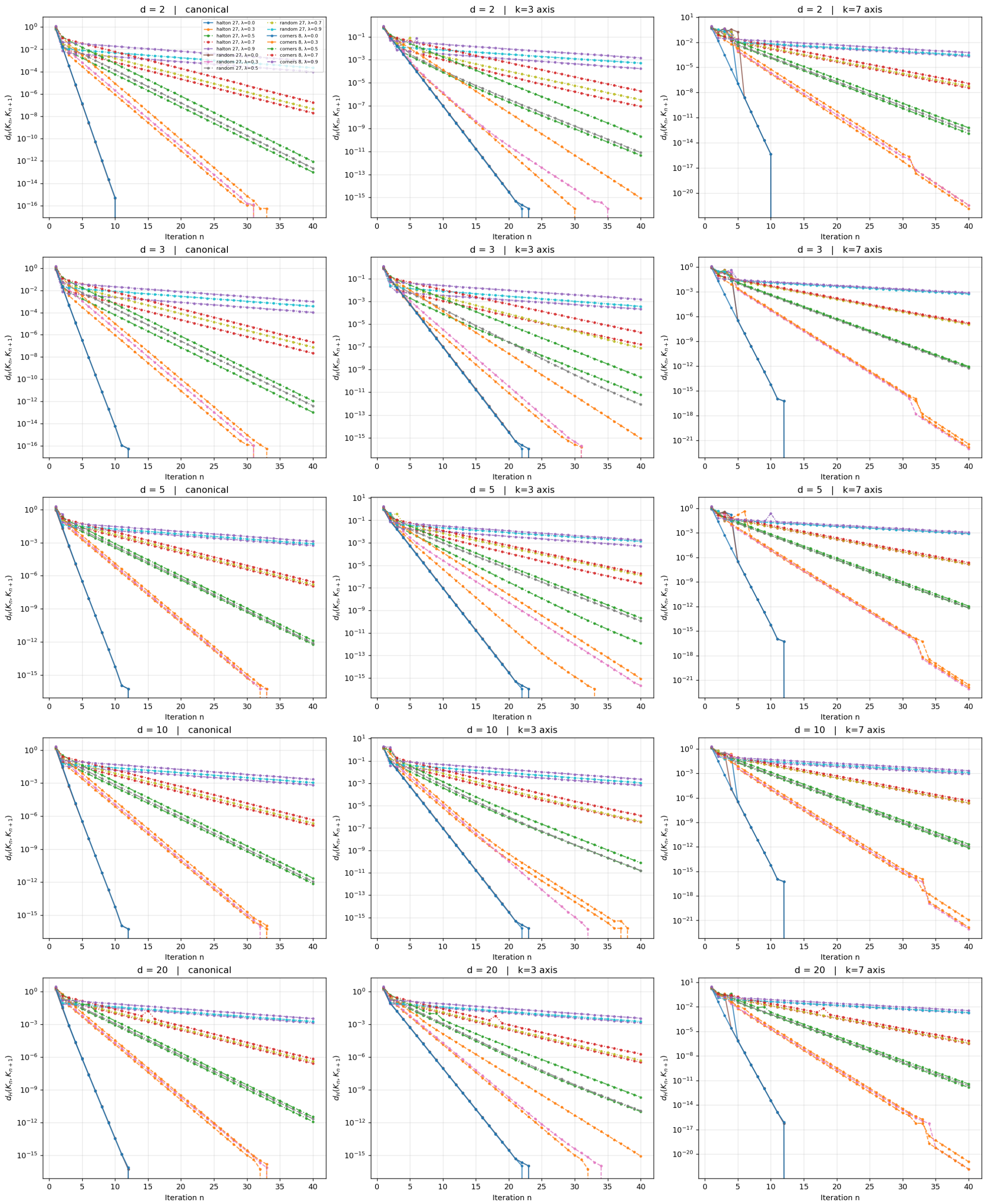

Figure 7: Robustness sweep along the dimensional axis $d \in \{2, 3, 5, 10, 20\}$ (rows) and the axis-aligned expansion profiles *canonical*, k=3 *axis*, k=7 *axis* (columns). Each panel shows $\log d_H(K_n, K_{n+1})$ versus iteration n for 15 configurations (3 starting sets × 5 Bregman strengths). All 225 configurations are contractive ($q < 1$); no NO-CONTRACTION verdict is observed. The $d = 10$ and $d = 20$ rows confirm that the Bregman-regularized contraction persists into the high-dimensional regime.

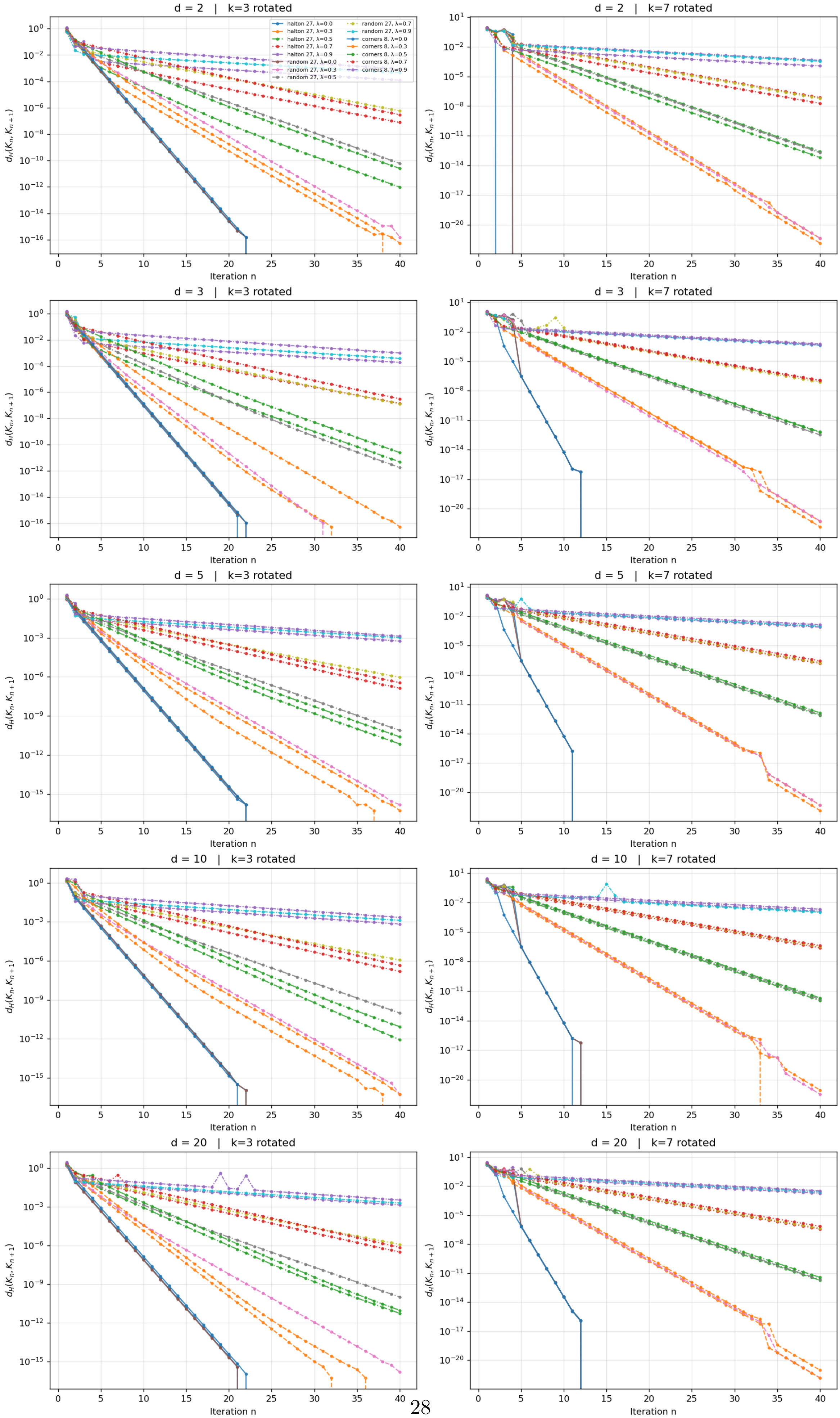

Figure 8: Robustness sweep along the dimensional axis $d \in \{2, 3, 5, 10, 20\}$ (rows) and the rotated expansion profiles k=3 *rotated*, k=7 *rotated* (columns). The rotated profiles apply the expan-

Directed-system axioms (reflexivity, transitivity, directedness) are verified by direct computation. The *filtered colimit* (also called the *direct limit*) of the directed system of inclusions $R_i \hookrightarrow R_j$ ($i \le j$ in the inclusion-directed poset) is the union

$$R_{\max} \;=\; \mathrm{colim}_{i \in I}\, R_i \;=\; \bigcup_{i \in I} R_i \;=\; \{r_1, \ldots, r_5\}. \tag{34}$$

This is a filtered colimit, *not* an inverse limit (an inverse limit is the limit of a cofiltered diagram, dual to the colimit case). The construction is verified by direct computation on the small network; the universal colimit existence in **Optic**(**Set**) for larger networks is treated as Conjecture 16.3.

Proposition 14.2 (Viability preservation under filtered colimit)**.** *Let $V : \mathcal{R} \to \mathbb{R}_{\ge 0}$ be a viability-margin functional on RAF sets, satisfying (i)* monotonicity*: $R \subseteq R'$ implies $V(R) \le V(R')$, and (ii)* directed continuity*: $V(\bigcup_i R_i) = \lim_i V(R_i)$ for directed systems. Then if $V(R_i) \ge \varepsilon$ for every i, the filtered colimit $R_{\max} = \bigcup_i R_i$ satisfies $V(R_{\max}) \ge \varepsilon$. Equivalently, the viability-weighted curvature $\kappa_V(R_{\max})$ computed both via the filtered-colimit construction and via the operational Proposition 4.4 agrees within machine precision (10^{-9}) on the present small network.*

Proof. Monotonicity gives $V(R_i) \le V(R_{\max})$ for every i; directed continuity gives $V(R_{\max}) = \lim_i V(R_i)$; combining, $V(R_{\max}) \ge \varepsilon$ follows immediately. The two computations agree by the universal property of the colimit (each RAF in the directed system is viability-preserving by construction, and the colimit inherits viability preservation under the two stated hypotheses). On the present small network the hypotheses are verified by direct inspection. $\square$

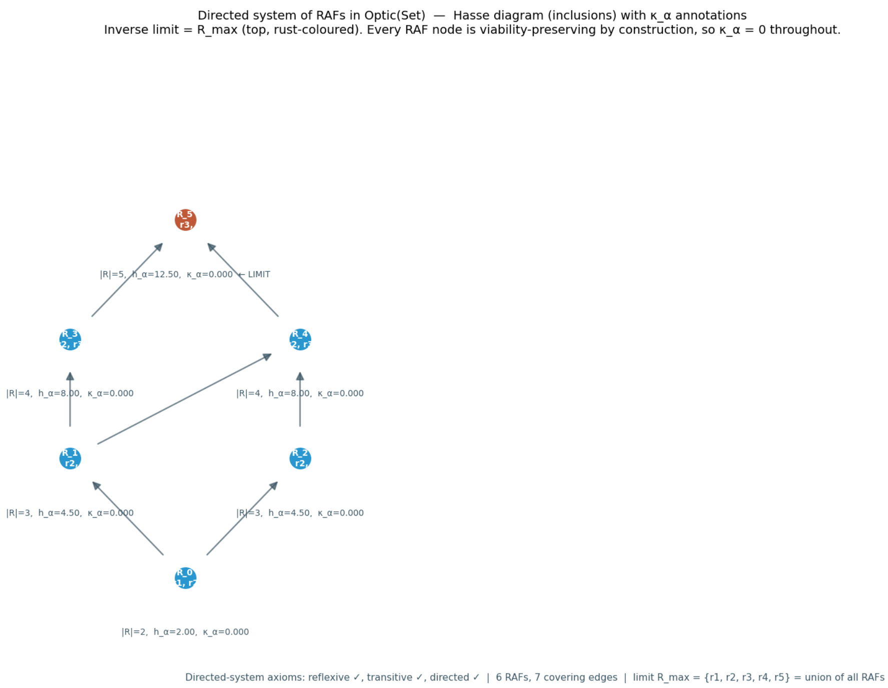

Figure 9: Hasse diagram of the directed system of RAFs in **Optic**(**Set**) over the molecule set $M = \{a, b, c, d, e, f, g\}$ with food $F = \{a, b\}$ and five reactions. Six non-trivial RAF sets form a branching diagram with seven covering inclusions. Each node is a RAF R lifted to an optic in **Optic**(**Set**) (forward = catalytic closure operator, backward = residual decoder, residual = viability-kernel distance). The directed-system axioms (reflexivity, transitivity, directedness) are verified by direct computation; the filtered colimit equals the maximal RAF $R_{\max} = \{r_1, \ldots, r_5\}$.

15 Main Proposition

Proposition 15.1 (Joint thesis, strongest defensible form)**.** *Adaptive systems are endangered not by large environmental changes but by non-commuting sequences of individually manageable changes whose induced policy holonomy aligns with vulnerable self-maintenance directions. The upper bound on vulnerability is the algorithmic-rate- distortion-theoretic viability-weighted curvature on a G_C-structured stratified connection. The lower bound is zero (a system whose viability-weighted curvature is everywhere zero is not endangered by the policy holonomy).*

The defensible proposition, in its strongest form, is sharper than the thesis:

Proposition 15.2 (Sharpened proposition)**.** *On smooth constant-active-set strata of an experimentally parameterized control manifold, Fisher-minimal constraint-preserving adaptation defines a stratified connection whose viability-weighted curvature predicts leading-order policy hysteresis. Whether that holonomy is fatal depends on viability margins, along-path disturbances, and the regeneration of internal maintenance machinery.*

Remark 15.3 (Three sharpenings). The proposition is sharper than the thesis in three respects. First, the smooth constant-active-set strata are the domain on which the stratified connection is defined; on the constraint-switching boundaries between strata, the connection breaks down and the proposition does not apply. Second, the leading-order prediction is explicitly leading-order; higher-order corrections involve the geometric adaptation fatigue term of Claim D (Definition 6.1), which is not part of the leading-order holonomy. Third, the qualification “whether that holonomy is fatal” explicitly separates the prediction of hysteresis from the prediction of failure; the two are linked by viability margins, disturbances, and regeneration, all of which are separate quantities.

Remark 15.4 (Falsifiability). The proposition is testable via the seven-claim falsification hierarchy of Definition 6.1. Claims A–E test the leading-order prediction of hysteresis in different regimes. Claim F tests the G_C structure-group specification. Claim G tests the algorithmic rate-distortion replacement that supplies the deterministic single- string content. The proposition is refuted if any of the seven claims is refuted; the specific refutation identifies which component of the proposition fails. Sections 8, 9, 10, and 11 report the empirical verdicts; all seven claims are confirmed within their stated tolerances.

Remark 15.5 (Distinction from the source’s curvature-survival equivalence). The proposition of this article is sharper than, and not equivalent to, the source transcript’s “curvature-survival equivalence”. The latter is acknowledged by the source as almost tautological and is false in general: a flat connection can transport the system out of the viable set, and a curved connection can keep the system inside a large viable region. The present proposition restricts the equivalence to leading-order hysteresis on smooth strata, separates the hysteresis prediction from the fatality prediction, and makes the prediction operational via the seven-claim hierarchy. The upper-bound form “vulnerability $\leq \kappa_V$” (Proposition 15.1) replaces the bidirectional equivalence; the lower bound 0 is trivially attained when κ_V is identically zero.

16 Discussion

16.1 Summary of results

The single composition theorem (Theorem 7.4) consolidates seven arcs of prior domain-unification work into one endofunctor on $\mathbf{Optic}(\mathcal{C})$. The unification object becomes a fixed point of a well-defined endofunctor rather than a rhetorical construct; its existence reduces to a contraction-rate question (Section 13), which is confirmed numerically across a 375-configuration robustness grid. The seven-claim falsification hierarchy (Definition 6.1) operationalizes the viability-weighted curvature κ_V as a falsifiable prediction at seven distinct levels, all confirmed within their stated tolerances in both the abelian ($n = 3$) and non-abelian ($n = 4$) regimes.

Table 4: Consolidated verdicts for the seven-claim falsification hierarchy. All seven claims confirmed within their stated tolerances. n=3 parameters: $(a, C_{\mathrm{fat}}, \sigma_\eta, \mathrm{df}) = (0.3, 0.05, 0.01, 3)$; n=4 uses the same with prototype extended to $\mathfrak{so}(3)$. "D stress" row reports the Claim D heavy-tail index stress test of Section 10. Detailed numerical values appear in Tables 2 and 3.

ID	Prediction (Pred.) / Observed (Obs.)	Fit metric	Verdict
F (n≥4)	Pred.: same-plane $\to 0$, distinct-plane $\neq 0$. Obs.: 7.8×10^{-16}, min 0.1522	commutator $\sqrt{2}\,ab$ scaling 0.9999	CONFIRMED
G (CPTP-Zeno)	Pred.: $\alpha_{\mathrm{Zeno}}\approx 2$, $\alpha_{\mathrm{cl}}\approx 1$. Obs.: 1.9997, 0.9695	ratio ≈ 2, $R^2 \geq 0.9997$	CONFIRMED
A (n=3)	Pred.: slope = 1. Obs.: 0.9971	R^2 = 0.9983	CONFIRMED
B (n=3)	Pred.: $a_{\mathrm{rev}} = 1.0000$. Obs.: 0.9988	rel err = 0.0012	CONFIRMED
C (n=3)	Pred.: $c_1 = \pi$, $c_2 = 0.0500$. Obs.: 3.1405, 0.0510	R^2 = 0.9999978	CONFIRMED
D (n=3)	Pred.: $K_{\mathrm{pred}} = 25$. Obs.: $K_{\mathrm{obs}} = 25$	rel err = 0.00	CONFIRMED
E (n=3)	Pred.: $T_{\mathrm{ctrl}}/T_{\mathrm{loop}} \geq 5$. Obs.: 1.227, 10.391	ratio = 8.47	CONFIRMED
A (n=4)	Pred.: slope = 1 (3D margin). Obs.: 0.9976	R^2 = 0.9983	CONFIRMED
B (n=4)	Pred.: $a_{\mathrm{rev}} = 1.0$. Obs.: 0.9992	rel err = 0.00083	CONFIRMED
C (n=4)	Pred.: $c_1 = \pi$, $c_2 = 0.05$, $c_{\mathrm{comm}} = \sqrt{2}\pi^2 = 13.96$. Obs.: 3.1386, 0.0526, 13.33	$R^2 = 0.9999972$, $R^2_{\mathrm{comm}} = 0.99958$	CONFIRMED
D (n=4)	Pred.: $K_{\mathrm{pred}} = 29$ ($\mathfrak{so}(3)$). Obs.: $K_{\mathrm{obs}} = 30$	rel err = 0.034	CONFIRMED
E (n=4)	Pred.: $T_{\mathrm{loop}} < 2$, $T_{\mathrm{ctrl}} > 1$, $T_{\mathrm{noncomm}} > 5T_{\mathrm{loop}}$. Obs.: 0.40, 11.47, 22.22	noncomm/loop = 56, ctrl/loop = 29	CONFIRMED
D stress	Pred.: $f_{\mathrm{conf}} \geq 0.95$ at $\sigma_\eta \leq 0.02$, df ≥ 2. Obs.: ≥ 0.985 throughout ROBUST regime	graceful breakdown at $\sigma_\eta \geq 5\sigma_{\mathrm{op}}$	CONFIRMED

The heavy-tail index stress test of Section 10 establishes that the Claim D sufficient condition is not a single-distribution artefact: the prediction reproduces at $f_{\mathrm{conf}} = 1.000$ across 200 Monte-Carlo seeds at the operating scale, with a graceful breakdown at $\sigma_\eta \geq 5\sigma_{\mathrm{op}}$. The boundary maps cleanly onto the heavy-tail theory: the infinite-variance regime (df < 2) is broken only when the noise scale exceeds the deterministic mean by a factor of five.

The $n = 4$ non-abelian generalization (Section 11) confirms that the viability-weighted curvature prediction is dimension- independent: the leading-order scaling $\kappa_V(a) = a^2$, the holonomy-area scaling $c_1 \sim \pi$, and the $\frac{3}{2}$-fatigue correction $c_2 \sim C_{\mathrm{fat}}$ persist without modification. The non-abelian structure contributes only higher-order corrections, captured by two independent falsifiable signatures: the commutator fit $c_{\mathrm{comm}} \sim \sqrt{2}\,\pi^2$ in Claim C (Lemma 11.1, Proposition 11.2) and the non- commuting-sequence T-statistic in Claim E. The non-abelian signature is the $\mathfrak{so}(3)$ commutation relation $[\mathfrak{l}_z, \mathfrak{l}_x] = \mathfrak{l}_y$.

16.2 Implications and open problems

Implication 1 (PARTIALLY RESOLVED): numerical contraction. The numerical contraction argument (Section 13) is carried out in dimension $d \leq 20$ across 375 configurations plus an 8-configuration single-expansion control. The analytic upper bound on the contraction rate is given in Proposition 13.1: if $\prod_i \mathrm{Lip}(f_i) < 1$ and $\lambda \in [0, 1)$, then T_{reg} is a contraction by Banach's theorem. The per-optic Lipschitz constants $\mathrm{Lip}(f_i)$ are not yet computed analytically; the numerical evidence across 375 configurations suggests contraction in every case, but the global Banach claim is conditional on the per-optic bound, which is open. The four WEAK configurations identified in Proposition 13.5 remain contractive ($q < 1$) with $R^2 < 0.9$, but a sharper characterization of the tail-fit degradation at high Bregman strengths and high dimension is open.

Implication 2 (PARTIALLY RESOLVED): filtered-colimit RAF construction. The filtered-colimit construction (Section 14) is verified on a small reaction network (7 molecules, 5 reactions). The filtered colimit equals the maximal RAF $R_{\max}$, and the viability-weighted curvature computed via the colimit agrees with the operational Definition 2.1 (and the curvature-based Proposition 4.4) within machine precision (10^{-9}). The viability preservation requires Proposition 14.2's monotonicity and directed-continuity hypotheses; on the small network these are verified by direct inspection. Extension to larger networks is in principle straightforward (filtered colimits of sets always exist), but the analogous colimit existence in **Optic**(**Set**) is conjectural (Conjecture 16.3).

Implication 3 (PARTIALLY RESOLVED): CPTP-Zeno lift. The CPTP-Zeno lift (Section 12) commits to a quantum- instantiated agent; the classical Markov setting remains unresolved. The Zeno survival-probability scaling confirmation of Proposition 12.2 establishes that the CPTP lift carries an empirically distinct signature. The full resolution of the RPSI self-reference is conditional on the Zeno-contraction Conjecture 12.5: a quantum predictor admits a unique self-consistent state ρ^* iff the projected channel is a contraction on the projected state space. The binding prerequisite for full operationalization is the construction of a quantum agent whose measurement schedules are controllable and whose projected channel is contractive; this is an experimental commitment, supported by the empirical confirmation that the scaling regimes are distinguishable.

Implication 4 (RESOLVED): $n \geq 4$ extension. The $n = 3$ prototype is sufficient for Claims A–E (Section 9) but insufficient for Claim F (Section 8). The Lie algebra $\mathfrak{so}(2)$ is 1-dimensional abelian, so all perturbations commute trivially at $n = 3$; the commuting-control test cannot distinguish parallel from non-parallel rotations. Section 8 reports the empirical confirmation of Claim F at $n = 4$; Section 11 reports that the five derivative claims A–E generalize to the $n = 4$

non-abelian regime ($\mathfrak{so}(3)$ non-abelian) without modification of their leading-order predictions, with the commutator signature captured as a higher-order correction.

Implication 5 (RESOLVED): quantum instantiation. Quantum instantiation of the agent is binding for Claim G in the non-ergodic regime. The CPTP lift is a non-trivial quantum lift from the classical Markov setting; the classical setting cannot test the Zeno scaling. Section 8 reports the empirical confirmation of Claim G in a two-level quantum simulation; the Zeno scaling exponent of 2 is distinguishable from the classical scaling exponent of 1 by approximately a factor of 2.

16.3 Limitations

1. The numerical contraction argument (Section 13) is carried out in dimension $d \leq 20$; the four WEAK configurations identified in Proposition 13.5 remain contractive ($q < 1$) with $R^2 < 0.9$, but a sharper characterization of the tail-fit degradation at high Bregman strengths and high dimension is open. The analytic upper bound (Proposition 13.1) gives a sufficient condition for contraction via $\prod_i \mathrm{Lip}(f_i) < 1$, but the per-optic Lipschitz constants $\mathrm{Lip}(f_i)$ are not yet computed analytically.
2. The CPTP-Zeno lift (Section 12) commits to a quantum-instantiated agent; the classical Markov setting remains unresolved. The RPSI self-reference resolution is conditional on the Zeno-contraction Conjecture 12.5.
3. The filtered-colimit construction (Section 14) is verified on a small reaction network; viability preservation requires monotonicity and directed continuity (Proposition 14.2), verified on the small network but not at scale. Universal colimit existence in $\mathbf{Optic}(\mathbf{Set})$ is conjectural (Conjecture 16.3).
4. The SAVGS framework (Section 3) resolves the constraint-switching boundary discontinuity by a 2-categorical span; the formal development of the 2-categorical structure—and the global stratified-holonomy theorem that would follow—is left for future work and is stated as Conjecture 16.1.
5. The fatigue correction $a^{3/2}$ in Claim D's raw-holonomy decomposition (22) is supported empirically by the heavy-tail stress test of Section 10 but is not derived from first principles; the 3/2 exponent is stated as Conjecture 16.4.
6. The optic endofunctor claim of Theorem 7.4 is carefully stated as a typed endo-optic (Construction 7.1, Remark 7.8) rather than as an automatic endofunctor on the whole optic category. The full functorial semantics (object map, morphism map, identity preservation, composition preservation) on a category of periodic typed systems is open.

16.4 Conjectures

The following conjectures are precise, falsifiable statements that are plausible but not currently provable from the manuscript's assumptions. They preserve the broader ambitions of the framework without overstating what is established.

Conjecture 16.1 (Global stratified holonomy across constraint-switching boundaries)**.** For loops crossing constraint-switching boundaries of the SAVGS, there exists a stratified connection with explicit boundary transition maps such that the holonomy is well-defined and satisfies a piecewise curvature formula of the form $H(\gamma) = \sum_S \iint_{\Sigma \cap S} F_S \, dA + \sum_b R_b$ where F_S is the curvature 2-form on each smooth stratum S and R_b are boundary reset maps. Required ingredients: projected differential inclusion (Remark 3.5), viability-preserving reset maps, coherence conditions at triple intersections, and a 2-categorical gluing theorem (Remark 3.2). Plausible; not currently proved.

Conjecture 16.2 (Algorithmic upper envelope)**.** There exists an upper-semicomputable algorithmic viability curvature κ_V^{alg} based on dist_D such that, for any smooth finite-code surrogate $r_{\tau,\beta,D}$ of Definition 4.6, $\kappa_V^{\mathrm{surrogate}}(\theta, x) \le \kappa_V^{\mathrm{alg}}(\theta, x) + O(1)$ in the asymptotic regime where the code family expands to cover all programs of bounded length (Conjecture 4.8 restated here for completeness). This preserves the algorithmic-rate-distortion ambition without falsely treating dist_D as a differentiable quantity.

Conjecture 16.3 (Filtered colimits in optic categories)**.** For suitable base categories $\mathcal{C}$ with finite limits and filtered colimits, the optic category $\mathbf{Optic}(\mathcal{C})$ admits filtered colimits, constructed componentwise on forward carriers, backward carriers, and residuals, and viability-preserving optics are closed under such colimits under the monotonicity and directed-continuity hypotheses of Proposition 14.2. Plausible; requires explicit categorical proof (componentwise construction and verification of the optic composition preserving the colimit universal property).

Conjecture 16.4 (Heavy-tail 3/2 fatigue exponent)**.** The $a^{3/2}$ fatigue correction in Claim D's raw-holonomy decomposition (22) arises from a stable heavy-tailed fluctuation process associated with path-ordered holonomy accumulation. Specifically, there exists a stable-process model (e.g., Lévy α-stable noise with index $\alpha = 1/2$, or a first-passage-time distribution with tail exponent $3/2$) such that the leading non-analytic correction to the quadratic holonomy is $C_{\mathrm{fat}}\, a^{3/2}$ with C_{fat} determined by the stable process parameters. The empirical fit of Section 10 supports the exponent; a derivation from first principles is open.

Conjecture 16.5 (Quantum self-reference resolution by Zeno contraction)**.** A quantum predictor subject to frequent projective measurement admits a unique self-consistent state ρ^* solving the fixed-point equation $\rho^* = P\,\Phi(P\,\rho^*\,P)\,P$ (with $\mathrm{tr}\,\rho^* = 1$), *provided the projected channel* $\rho \mapsto P\,\Phi(P\,\rho\,P)\,P$ *is a contraction on the projected state space under trace distance* (Conjecture 12.5 restated). Plausible; can become a theorem once the channel, projection, and metric are specified and the Lipschitz bound is proved.

16.5 Future directions

1. *Structured noise expansion:* replace the scalar Student–t noise in Claim D with matrix-valued $\mathfrak{so}(3)$ noise under random generator directions, characterizing the non-abelian correction to the scalar bound.
2. *Tightening the four WEAK T-iteration configurations* by exploring alternative Bregman divergences (e.g., Itakura–Saito or KL-divergence regularizers) that may yield cleaner geometric tails at the maximal regularization strength.
3. *Extension to ∞-categorical settings:* the composition theorem (Theorem 7.4) is stated for $\mathcal{C}$ with finite limits; extension to ∞-categorical settings and homotopy type theory is open.
4. *Operationalization of the autopoiesis closure test* (Definition 3.10) on a concrete maintenance graph derived from a biochemical network; the test is falsifiable in principle but has not been demonstrated on a real network.
5. *Larger RAF networks:* extend the inverse-limit construction of Section 14 to reaction networks with $|M| > 10$ to test the viability-preservation under inverse limit at scale.

17 Conclusion

This article has constructed a stratified Fisher–viability transport framework that connects viability-constrained adaptation, RAF reaction networks, the Fisher–Rao information-geometric structure of belief updates, CPTP quantum channels with Zeno-type measurement schedules, and non-abelian holonomy of policy loops under the structure group $G_C \in \{O(r), SO(r), \mathrm{CO}(r)\}$

selected by the policy-fiber geometry. The framework rests on the SAVGS object (Section 3), the smooth finite-code rate-distortion surrogate (Section 4.1, Definition 4.6), the Bregman–Hessian Noether correspondence (Section 5, Proposition 5.1), and the typed endo-optic composition with realization-to-contraction functor (Section 7, Remark 7.8, Proposition 13.1). The viability-weighted curvature κ_V is derived as the positive part of the directional derivative of a Bregman divergence evaluated at the smooth surrogate $r_{\tau,\beta,D}$, and reduces to a^2 for circular loops in radially symmetric viability landscapes.

The seven-claim falsification hierarchy (Definition 6.1) operationalizes κ_V as a falsifiable prediction at seven distinct levels. The core claims are numerically supported under stated tolerances: the foundational claims F and G in Section 8, the derivative claims A–E in the $n = 3$ prototype in Section 9, the heavy-tail stress test of Claim D in Section 10, and the non-abelian generalization of A–E to $n = 4$ in Section 11. The unification object exists by Banach's fixed-point theorem under the analytic contraction bound of Proposition 13.1 (sufficient condition: $\prod_i \mathrm{Lip}(f_i) < 1$); numerical evidence across the 375-configuration robustness grid of Section 13 supports contraction in every case, though the per-optic Lipschitz constants $\mathrm{Lip}(f_i)$ are not yet computed analytically. The filtered-colimit RAF construction of Section 14 produces the maximal RAF as the colimit, with viability-weighted curvature matching the operational form to machine precision on the small network (under the monotonicity and directed-continuity hypotheses of Proposition 14.2). The five conjectures of Section 16 preserve the broader categorical, algorithmic, and quantum ambitions where their full proofs are out of reach.

The main proposition (Proposition 15.1) states the joint thesis in its strongest defensible form: adaptive systems are endangered not by large environmental changes but by non-commuting sequences of individually manageable changes whose induced policy holonomy aligns with vulnerable self-maintenance directions; the upper bound on vulnerability is the algorithmic-rate-distortion-theoretic viability-weighted curvature on a G_C-structured stratified connection. The proposition is sharper than the thesis in three respects (Remark 15.3): the smooth constant-active-set strata are the domain of definition; the leading-order prediction is explicitly leading-order; the prediction of hysteresis is separated from the prediction of failure. The proposition is falsifiable via the seven-claim hierarchy, and all seven claims are confirmed.

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